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The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM11.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Section 3, page 75: the proofs that a radial drawing exists, is 2-cell, and has a unique edge of Γ in each region are added.
2. **Statement mistake.** Statement of (3.1), page 76: a local hypothesis on X and $X \cap Y$ is added.
3. **Statement mistake.** Statement of (6.5), page 92: the "if" part assumes that $\mathscr{T}$ has order $\neq 2$ or that Γ has no loop.
4. **Statement mistake.** Statement of (6.6), page 92: for $\theta = 2$, neither Γ nor Γ^* may have a loop.

[p. 72] Graph Minors. XI. Circuits on a Surface

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Let Γ be a graph or hypergraph drawn on a connected surface which is not a sphere, in such a way that every non-null-homotopic curve meets the drawing at least θ times. We show that this defines a "tangle of order θ" in Γ, in the sense of earlier papers of this series. Also, there is a natural distance function defined by the drawing, and we show that for any point c of the surface, and any $\theta' < \theta$, the set of points within distance θ' of c is simply-connected.

1. Introduction

This is one of a series of papers establishing lemmas towards the proof of a theorem concerning the structure imposed on graphs by the exclusion of a fixed minor. The main proof of that theorem, which will appear in a later paper, involves among other things a long and cumbersome argument about graphs drawn on connected surfaces. Unfortunately, the case when the surface is a sphere is in some ways different from (and more difficult than) the higher genus case. It would be a pity to have to treat the two cases separately, because large parts of the arguments involved are identical, and the purpose of this paper is to develop a framework within which both cases can be handled simultaneously.

If the surface is not a sphere, and the drawing is "θ-representative," that is, every non-null-homotopic closed curve meets the drawing at least θ times, then any simple closed curve which meets the drawing $< \theta$ times bounds a unique disc, which we may call its "inside." The main problem with the sphere case is the lack of unique "insides," and we can overcome [p. 73] it by the use of "tangles," developed in earlier papers. Roughly, a tangle of order θ in a graph can be thought of as a highly coherent area of the graph, which clearly resides on one side or the other of every separation

*Part of this work was performed under a consulting agreement with Bellcore.

of order $< \theta$. If we have a graph drawn on the sphere with a tangle of order $\geq \theta$ then for any simple closed curve C which meets the drawing $< \theta$ times (and which meets the drawing only in vertices), the two discs it bounds define a separation of the graph of order $< \theta$, and we may define the "inside" of C to be whichever of these discs does not "include" the tangle. We find that, with this definition of "inside," we can handle drawings on the sphere just as easily as drawings on higher surfaces. Conveniently, there is a complementary result that, if a graph has a θ-representative drawing on a connected surface not the sphere, this defines a tangle in the graph of order θ; and this tangle resides "outside" every null-homotopic curve which meets the drawing $< \theta$ times, just as in the sphere case. This permits us to completely unify the sphere and non-sphere cases.

Our first objective in this paper is to prove that a graph with a θ-representative drawing on a connected surface not the sphere contains an associated tangle of order θ. This occupies Sections 5 and 6. (The earlier sections are concerned with definitions.) Secondly, given a graph on a surface (highly representative if the surface is not a sphere, and with a designated high order tangle if the surface is a sphere, so that in either case we have a definition of "inside"), there is associated a natural distance function on the surface, and in Sections 7-9 we study some of its properties. In particular, it turns out that for any point c of the surface, the set of all points within distance θ' of c (for any $\theta' < \theta$) is simply-connected, and thus every point has a "large" planar neighborhood. This will be useful in subsequent papers.

We are concerned with simple closed curves in the surface which meet the drawing Γ only in vertices, and only a small number of times. There is an associated graph (the "radial drawing" of Γ) in which such curves correspond to circuits, and it is most convenient to work in terms of this rather than directly in terms of Γ.

One can also draw hypergraphs on surfaces, and define tangles in hypergraphs, and it turns out that all our results generalize to hypergraphs. (The radial drawing of a drawing of a hypergraph is still a graph, not a hypergraph, and so this generalization to hypergraphs makes very little difference.) Thus we formulate the present results in terms of hypergraphs.

2. Hypergraphs and Tangles

A hypergraph G consists of a set $V(G)$ of vertices, a set $E(G)$ of edges, and an incidence relation between them. All hypergraphs in this paper are [p. 74] finite. If $e \in E(G)$, the vertices incident with e are the *ends* of e. If G, H are hypergraphs, H is a *subhypergraph* of G (written $H \subseteq G$) if

$V(H) \subseteq V(G)$, $E(H) \subseteq E(G)$, and every edge of H has the same ends in H as it has in G. If $H_1, H_2 \subseteq G$ we define $H_1 \cap H_2$ to be the subhypergraph of G with vertex set $V(H_1) \cap V(H_2)$ and edge set $E(H_1) \cap E(H_2)$, and we define $H_1 \cup H_2$ similarly. A *separation* of G is a pair (A, B) of subhypergraphs with $A \cup B = G$ and $E(A \cap B) = \emptyset$, and its *order* is $|V(A \cap B)|$.

A *tangle* of order $\theta \geq 1$ in G is a set $\mathscr{T}$ of separations of G, each of order $< \theta$, such that

(i) for every separation (A, B) of G of order $< \theta$, $\mathscr{T}$ contains one of (A, B), (B, A)

(ii) if $(A_1, B_1), (A_2, B_2), (A_3, B_3) \in \mathscr{T}$ then $A_1 \cup A_2 \cup A_3 \neq G$

(iii) if $(A, B) \in \mathscr{T}$ then $V(A) \neq V(G)$.

We refer to these as the *first, second and third tangle axioms.* Tangles were introduced in [4].

3. Surfaces, Drawings, and Homotopy

A *surface* is a compact 2-manifold, without boundary. (In this paper we are really only concerned with connected surfaces.) Let Σ be a surface. A *line* in Σ is a subset homeomorphic to $[0, 1]$, and its *ends* are defined in the natural way. An *O-arc* is a subset homeomorphic to a circle. If $\Delta \subseteq \Sigma$, then $\bar{\Delta}$ denotes the closure of Δ, and $bd(\Delta)$ denotes $\bar{\Delta} \cap \overline{\Sigma - \Delta}$. A *drawing* in Σ is a pair (U, V), where $V \subseteq U \subseteq \Sigma$, U is closed, V is finite, $U - V$ has only finitely many arc-wise connected components, called *edges*, and for every edge e, either $\bar{e}$ is a line whose set of ends is $\bar{e} \cap V$, or $\bar{e}$ is an O-arc and $|\bar{e} \cap V| = 1$. The components of $\Sigma - U$ are called *regions.* The members of V are called *vertices.* If $\Gamma = (U, V)$ is a drawing, we write $U = U(\Gamma)$, $V = V(\Gamma)$, and $E(\Gamma)$, $R(\Gamma)$ are defined to be the sets of edges and regions of Γ, respectively.

If v is a vertex and e is an edge or region of Γ, we say v is *incident with* e (and vice versa) if v is contained in the closure of e. If e is an edge and r is a region, we say e is *incident with* r (and vice versa) if $e \cap \bar{r} \neq \emptyset$. We say Γ' is a *subdrawing* of a drawing Γ if $V(\Gamma') \subseteq V(\Gamma)$ and $E(\Gamma') \subseteq E(\Gamma)$ (and hence $U(\Gamma') \subseteq U(\Gamma)$). If Γ' is a subdrawing of Γ we write $\Gamma' \subseteq \Gamma$. If Γ_1, Γ_2 are subdrawings of Γ, we define $\Gamma_1 \cup \Gamma_2$, $\Gamma_1 \cap \Gamma_2$ in the natural way. If Γ is a drawing, then $V(\Gamma)$, $E(\Gamma)$ and the incidence relation between them define a graph, and in this sense we may regard Γ as a graph, and use graph-theoretic terminology for drawings.

$\Delta \subseteq \Sigma$ is a *closed disc* if it is homeomorphic to $\{(x, y) : x^2 + y^2 \leq 1\}$, and [p. 75] an *open disc* is defined similarly. A *painting* Γ in a surface Σ is a pair (U, V), where $V \subseteq U \subseteq \Sigma$, V is finite, $U - V$ has only finitely many

arc-wise connected components, called *edges*, and for every edge e, $\bar{e}$ is a closed disc and $\bar{e} \cap V \subseteq bd(\bar{e})$. The components of $\Sigma - U$ are called *regions*, and further terminology is the same as for drawings. If Γ is a painting, then $V(\Gamma)$, $E(\Gamma)$ and the incidence relation between them define a hypergraph, and in this sense we may regard Γ as a hypergraph.

A painting or drawing Γ in Σ is *2-cell* if Σ is connected and every region is an open disc (and hence Γ is not null). Now we wish to define a "radial drawing" of a 2-cell painting or drawing. Roughly, we select a point from each region of Γ, and make it adjacent to each vertex of Γ incident with the region, thereby obtaining a bipartite drawing. However, a region may be "incident more than once" with the same vertex, and so we must be a little careful. Let Γ be a 2-cell painting or drawing in Σ, and let Γ' be a drawing in Σ. We say that Γ' is a *radial drawing* of Γ if it satisfies the following conditions:

(i) $U(\Gamma) \cap U(\Gamma') = V(\Gamma) \subseteq V(\Gamma')$

(ii) each region r of Γ contains a unique vertex, v_r say, of Γ'; let

$$W = \{v_r : r \in R(\Gamma)\}$$

(iii) Γ' is bipartite, and $(V(\Gamma), W)$ is a bipartition of Γ'

(iv) if e, f are edges of Γ' with the same ends v, v_r say, where $v \in V(\Gamma)$ and $r \in R(\Gamma)$, then $\bar{e} \cup \bar{f}$ does not bound a closed disc in $r \cup \{v\}$

(v) Γ' is maximal subject to (i), ..., (iv).

It is not difficult to show that every 2-cell painting or drawing Γ has a radial drawing Γ', that Γ' is 2-cell, and that every region of Γ' includes a unique edge of Γ, unless $|V(\Gamma)| = 1$ and $E(\Gamma) = \emptyset$. Moreover, a radial drawing is unique up to automorphisms of Σ which fix $U(\Gamma)$ pointwise. We ~~omit~~give the proofs of the first three of these assertions, which are elementary but rather tedious, and we omit the proof of the last one.

First let Γ be a drawing. At a vertex x of a drawing, the edges at x (a loop counted twice) leave x in a cyclic order. We call these ends of edges the *half-edges* at x, and the parts of a small neighbourhood of x between two consecutive half-edges the *corners* at x. Each corner is connected and misses the drawing, and so it lies in a region of the drawing. Its two *sides* are the half-edges that bound it. If x is in the closure of a region, then some corner at x lies in this region, since near x the points not in the drawing are in the corners. We use several times that a connected set that misses the boundary of a closed disc lies in its interior or outside it.

For the existence, choose a point in each region of Γ. These points and the vertices of Γ, with no edges, form a drawing that satisfies (i)–(iv). We show below that every drawing that satisfies (i)–(iv) has at most

$|V(\Gamma)| + 2|E(\Gamma)|$ edges. Then some drawing Γ' has the most edges among those that satisfy (i)–(iv), and it satisfies (v). For let Γ'' satisfy (i)–(iv) and contain Γ'. By (ii) and (iii), Γ'' has the same vertices as Γ', and it has no more edges, and so $\Gamma'' = \Gamma'$.

For the bound, let Γ' satisfy (i)–(iv), and let H be the drawing formed by Γ and Γ', which is a drawing by (i). By (i), each edge of Γ' misses $U(\Gamma)$, and so it lies in a region r of Γ. By (iii) it joins a vertex of Γ to a vertex of Γ' not in $V(\Gamma)$, which is in the closure of r but not in $U(\Gamma)$, and so is v_r by (ii). In particular each edge of Γ' has just one half-edge at a vertex of Γ. Let $x \in V(\Gamma)$, suppose that there are at least two half-edges of H at x, and suppose that both sides of a corner at x are half-edges of Γ'. The corner lies in a region r of Γ, and its sides belong to two different edges f, g of Γ' that join x to v_r. Choose points of f and of g near x, and join them by an arc through the corner. With the rest of f and g, this arc forms an O-arc in r, which bounds a closed disc in r, since r is an open disc. The part of the corner cut off by the arc contains x, which is not in r, and so it lies outside this disc. Together they form a closed disc in $r \cup \{x\}$ bounded by $\bar{f} \cup \bar{g}$, contrary to (iv). So the half-edge that follows a half-edge of Γ' in the cyclic order at x is a half-edge of Γ. Map each edge of Γ' to the half-edge that follows its half-edge at its end x in $V(\Gamma)$, or to x itself if there is only one half-edge at x. This map is one-to-one, since the map taking each half-edge at x to the next one is one-to-one, and since at most one edge of Γ' is at x if there is only one half-edge at x. Its values are among the vertices and the $2|E(\Gamma)|$ half-edges of Γ, which gives the bound.

Now let Γ' be a radial drawing of Γ, and let H be as before. Condition (v) is used in the following way. Let r be a region of Γ, let x be a vertex of Γ, and let K be a line in $r \cup \{x\}$ with ends v_r and x that meets $U(\Gamma')$ only in its ends. Then Γ' with K added as an edge satisfies (i)–(iii), and so by (v) it fails (iv). That is, K and the closure of some edge h of Γ' with ends x and v_r bound a closed disc in $r \cup \{x\}$. The interior of this disc contains no point of $U(H)$. For the interior lies in r, and so it misses $U(\Gamma)$, and it contains no vertex of Γ', since the only vertices of Γ' in $r \cup \{x\}$ are v_r and x. Let g be an edge of Γ' that meets the interior. It misses the boundary of the disc, since it misses K and h, and so it lies in the interior, and its ends are vertices of Γ' in the disc, which are x and v_r. Then g splits the disc into two closed discs, and one of them is bounded by $\bar{g} \cup \bar{h}$ and lies in $r \cup \{x\}$, contrary to (iv).

Now suppose that Γ has an edge. Every vertex x of Γ is an end of an edge of Γ. For otherwise a small closed disc around x meets $U(\Gamma)$ only in x. Its boundary bounds a closed disc in the region of Γ that contains it,

since this region is an open disc. These two closed discs have the same boundary and are different, since only one contains x, and so their union is open and closed in Σ, and it is Σ, since Σ is connected. So $U(\Gamma) = \{x\}$, a contradiction. Also the closure of each region r of Γ contains a vertex of Γ. For $r \neq \Sigma$, since $U(\Gamma) \neq \emptyset$, and so the frontier of r is not empty, since Σ is connected. It lies in $U(\Gamma)$, and it contains a vertex, or a point of an edge and then also the ends of this edge. We obtain the following facts, in order.

(a) *For each region r of Γ and each vertex x of Γ in the closure of r, some edge of Γ' joins x to v_r.* For $r - U(H)$ is the union of the finitely many regions of H inside r, and it is dense in r, since $U(H) \cap r$ consists of v_r and edges of Γ'. So x is in the closure of a region p of H inside r. Also v_r is in the closure of every region q of H inside r. For q is open and non-empty, and $q \neq r$, since $v_r \in r - q$. Since r is connected, it follows that the frontier of q meets r. A point of this frontier in r is in $U(\Gamma') \cap r$, which consists of v_r and edges of Γ' that have v_r in their closure, and if it is in such an edge then the whole edge, and so v_r, is in the closure of q. Hence some corner at x and some corner at v_r lie in p. Since p is open and connected, there is a line K as above, which runs from v_r through a corner in p, then inside p, and reaches x through a corner in p. Then h joins x to v_r.

(b) *At each vertex x of Γ, each corner has one side in Γ and the other in Γ'.* We saw in the proof of the bound that the two sides of a corner are not both in Γ', if there are at least two half-edges at x. If there is only one, it is a half-edge of Γ, since x is an end of an edge of Γ. So suppose that both sides of a corner at x are half-edges of Γ (the same one, if it is the only half-edge at x). The corner lies in a region p of H, inside a region r of Γ, and as in (a) there is a line K as above, with all points other than its ends in p, that reaches x inside this corner. Near x, the closed disc bounded by K and $\bar{h}$ lies on one side of K, between K and h in the cyclic order. Since its interior misses $U(H)$, no half-edge of H at x lies strictly between K and h on this side. So the half-edge of h at x is next to K, and it is one of the two sides of the corner, which are half-edges of Γ, a contradiction.

(c) *For each region r of Γ, different corners at v_r lie in different regions of H.* For suppose that two of them lie in one region p of H. Since p is open and connected, there is an O-arc J with $J - \{v_r\} \subseteq p$ that leaves v_r through one of these corners and returns through the other. Since $J \subseteq r$ and r is an open disc, J bounds a closed disc in r. No edge of Γ' meets the interior of this disc. For such an edge misses J, so it would lie in the disc with both its ends, while each edge of Γ' has an end in $V(\Gamma)$, outside r. But near v_r the disc lies on one side of J, and since the two corners are different, each side of J near v_r contains a half-edge of Γ' at v_r, a contradiction.

(d) *For each region p of H, one or two edges of Γ' are in the closure of p, and if two, each of them has different regions of H on its two sides.* Let r be the region of Γ that includes p. As in (a), v_r is in the closure of p, and by (a), v_r is an end of an edge of Γ', since the closure of r contains a vertex of Γ. So some corner at v_r lies in p, and by (c) just one does. Each edge of Γ' in the closure of p has v_r as an end, and the points of p near it close to v_r lie in a corner beside it, which is then the corner in p. So each such edge is a side of this corner. If there are two of them, then the two corners beside one of them at v_r are different, and so by (c) they lie in different regions of H.

(e) *The closure of each region p of H includes exactly one edge of Γ, and contains no vertex of Γ other than the ends of this edge.* Let a half-edge of an edge of Γ in the closure of p be given. Some corner beside it lies in p, and by (b) the other side of this corner is a half-edge of an edge of Γ', which is in the closure of p. Each edge of Γ' has one half-edge at a vertex of Γ, and at most two half-edges are next to it there, one on each side. If there are two edges of Γ' in the closure of p, then by (d) only one side of each of them lies in p. So in both cases of (d), at most two half-edges of Γ are next to these edges of Γ' with the corner between them in p. An edge of Γ in the closure of p gives two of them, its two half-edges, and so there is at most one such edge. There is at least one, since if f is an edge of Γ' in the closure of p, then at its end in $V(\Gamma)$ a corner beside f lies in p, and by (b) the other side of this corner is a half-edge of Γ. Finally, if a vertex of Γ is in the closure of p, then some corner at it lies in p, and by (b) one side of this corner is a half-edge of Γ, whose edge is the edge of Γ in the closure of p.

(f) *Each region s of Γ' includes exactly one edge e of Γ, and s is the union of e and the regions of H whose closure includes e.* This is the third assertion. Since $V(\Gamma) \subseteq U(\Gamma')$, the set s is the union of the edges of Γ in s and the regions of H in s. By (e), the closure of each of these regions includes exactly one edge of Γ, and this edge lies in s, since it misses $U(\Gamma')$ and meets the closure of s. In particular s includes an edge of Γ. For each edge f of Γ in s, let s_f be the union of f and the regions of H whose closure includes f. It is open, since each point of f has a neighbourhood that meets $U(H)$ only in f, and whose points off f lie in regions of H beside f. The sets s_f cover s, and they are disjoint by (e). Since s is connected, there is only one of them.

(g) *Γ' is 2-cell, whether or not Γ has an edge.* First, for each region r of Γ, v_r is an end of an edge of Γ'. If Γ has an edge, this follows from (a), since the closure of r contains a vertex of Γ. Otherwise $V(\Gamma) \neq \emptyset$, since Γ is 2-cell and so not null. If no edge of Γ' ends at v_r, then $U(\Gamma')$ meets r

only in v_r, and a line K as above, from v_r to a vertex of Γ on the frontier of r with its other points in r, gives an edge h at v_r, a contradiction. Next, each region p of H is an open disc. For p lies in a region r of Γ, which is an open disc. An O-arc F in p bounds a closed disc Δ in r. Each point of $U(\Gamma')$ in r lies in the closure of an edge of Γ' with an end outside r. This closure is connected, misses F, and has a point outside Δ, and so it misses Δ. Hence Δ misses $U(H)$, and so $\Delta \subseteq p$. Thus p is a connected open subset of the open disc r in which every O-arc bounds a closed disc, and so p is an open disc, since a connected open subset of the plane with this property is an open disc. If Γ has no edge, then $H = \Gamma'$, and we are done.

Otherwise, each edge e of Γ has two different regions of H beside it. For suppose that the two corners beside e at an end x of e lie in one region p of H. By (b), their other sides are half-edges of edges f_1, f_2 of Γ', which are in the closure of p. Let y be the other end of e ($y = x$ if e is a loop). As in (e), the half-edge of e at y is next to the half-edge of an edge f of Γ' in the closure of p, with the corner between them in p. Suppose first that $f_1 \neq f_2$. Then by (d) these are the only edges of Γ' in the closure of p, and each has different regions of H on its two sides. So f is f_1 or f_2, and since an edge of Γ' has just one half-edge at a vertex of Γ, $y = x$, and e is a loop. But then the half-edge of f at x is next to both half-edges of e at x, with the corners between them in p, and so p is on both sides of f, contrary to (d). Now suppose that $f_1 = f_2$. Then x has just two half-edges, one of e and one of f_1, and p is on both sides of f_1. By (d), f_1 is the only edge of Γ' in the closure of p, and so $f = f_1$, and as before $y = x$. Then e is a loop at x, and x has three half-edges, a contradiction. By (f), it follows that each region of Γ' is the union of an edge e of Γ and the two regions of H beside e, which are disjoint open discs whose closures include e, and so it is an open disc, since gluing two disjoint open discs along an edge in the frontier of both gives an open disc. So Γ' is 2-cell.

Now let Γ be a painting. If the closure $\bar{e}$ of some edge e of Γ contains no vertex, then e is the only edge and Γ has no vertex. For the sets $bd(\bar{f})$ ($f \in E(\Gamma)$), with $V(\Gamma)$ and one more point on each $bd(\bar{f})$ as vertices, form a drawing whose regions are the regions of Γ and the sets $\bar{f} - bd(\bar{f})$ ($f \in E(\Gamma)$), which are open discs. The point set of this drawing is connected. For the frontier of an open disc in a surface is connected, and the frontier of each region is not empty, since Σ is connected. So if the point set were the union of two disjoint non-empty closed sets C_1, C_2, the frontier of each region would lie in C_1 or in C_2, and adding to each C_i the closures of the regions whose frontier lies in C_i would split Σ into two disjoint non-empty closed sets, which is impossible. But $bd(\bar{e})$ is closed and disjoint from the rest of

this point set, which is also closed, and so the rest is empty. In this case Σ is a sphere, since it is the union of the closed disc $\bar{e}$ and the region $\Sigma - \bar{e}$ of Γ, which is an open disc. By (i) and (iii), Γ' has no edge, since $V(\Gamma) = \emptyset$, and so it has one vertex. So the only region of Γ' is Σ minus a point, which is an open disc, and it includes e.

Otherwise let D be the drawing formed by the sets $bd(\bar{e})$ ($e \in E(\Gamma)$), with vertex set $V(\Gamma)$. Its regions are the regions of Γ and the sets $\bar{e} - bd(\bar{e})$ ($e \in E(\Gamma)$), since these sets are connected, open and disjoint, and their union is $\Sigma - U(D)$. So D is 2-cell. Let Γ' satisfy (i)–(iv) for Γ. For each edge e of Γ, add to Γ' a new vertex in $\bar{e} - bd(\bar{e})$, joined by one edge inside $\bar{e}$ to each vertex of Γ in $\bar{e}$. This gives a drawing G that satisfies (i)–(iv) for D. For (i)–(iii) hold by the construction, and (iv) holds for the new edges since no two of them have the same ends, and for the other edges since the regions of Γ are regions of D. Since G has at least as many edges as Γ', the bound for D bounds the number of edges of Γ', and the existence follows as before.

Now let Γ' be a radial drawing of Γ. Then G is a radial drawing of D. For let G'' satisfy (i)–(iv) for D and contain G. The edges of G'' in the regions of Γ, with the vertices of Γ', form a drawing that satisfies (i)–(iv) for Γ and contains Γ', and so by (v) they are the edges of Γ'. An edge of G'' in $\bar{e} - bd(\bar{e})$ joins the new vertex in it to a vertex x of Γ in $\bar{e}$. If two such edges had the same ends, they would form an O-arc in $\bar{e}$ meeting $bd(\bar{e})$ only in x, which bounds a closed disc in $(\bar{e} - bd(\bar{e})) \cup \{x\}$, contrary to (iv). So $G'' = G$. By the case of drawings, each region of G includes a unique edge of D, and each region of the drawing formed by D and G is an open disc. Let s be a region of Γ'. Each edge of Γ that meets s lies in s, since it misses $U(\Gamma')$ by (i), and each region of G that meets s lies in s, since $U(\Gamma') \subseteq U(G)$. Since $V(\Gamma) \subseteq U(\Gamma')$, the set s is the union of the edges of Γ in s and the regions of G in s. For each edge e of Γ in s, the union of e and the regions of G that include an edge of D in e is open, as in (f). These sets cover s, and they are disjoint, since each region of G includes just one edge of D. Since s is connected, there is only one of them. So s includes a unique edge e of Γ. Moreover, each edge a of D in e has two regions of the drawing formed by D and G beside it, by (g) for D and G. Each of these regions lies inside $\bar{e}$ or misses $\bar{e}$, since it is connected and misses $bd(\bar{e})$. One is inside and one is outside, since near a point of a one side of a is inside $\bar{e}$ and the other is outside. The regions outside, for different edges a, are different, by (e) for D and G. So s is formed by the open disc $\bar{e} - bd(\bar{e})$ and, for each edge a of D in e, the edge a and the region beside it outside $\bar{e}$. Adding these one at a time keeps an open disc, since each time we glue an open disc, disjoint from

the set built so far, along the edge a. So Γ' is 2-cell.[F1]

A curve in a surface Σ is a continuous function $\phi : [0,1] \to \Sigma$. For any curve ϕ, $|\phi|$ denotes $\{\phi(x) : 0 \le x \le 1\}$. A curve ϕ is *constant* if $|\phi|$ is a singleton. Now let $Z \subseteq \Sigma$, and let ϕ_0, ϕ_1 be curves. We say ϕ_0 is *homotopic to* ϕ_1 in Z if there is a continuous function $\psi : [0,1] \times [0,1] \to Z$ such that

$$\begin{aligned}
\psi(x,0) &= \phi_0(x) && (0 \le x \le 1)\\
\psi(x,1) &= \phi_1(x) && (0 \le x \le 1)\\
\psi(0,y) &= \phi_0(0) = \phi_1(0) && (0 \le y \le 1)\\
\psi(1,y) &= \phi_0(1) = \phi_1(1) && (0 \le y \le 1).
\end{aligned}$$

In particular, it is necessary that $\phi_0(0) = \phi_1(0)$ and $\phi_0(1) = \phi_1(1)$. We shall assume without proof certain well-known results about homotopy, such as [p. 76] that it is an equivalence relation. A curve is *null-homotopic in* Z if it is homotopic to a constant curve in Z.

Similarly, let S^1 be the circle $\{(x,y) : x^2 + y^2 = 1\}$. For $\phi\colon S^1 \to \Sigma$, we define $|\phi| = \{\phi(x) : x \in S^1\}$. If $Z \subseteq \Sigma$ and $\phi_0, \phi_1\colon S^1 \to \Sigma$ are continuous, we say ϕ_0 is *homotopic to* ϕ_1 *in* Z if there is a continuous function $\psi\colon S^1 \times [0,1] \to Z$ such that

$$\begin{aligned}
\psi(x,0) &= \phi_0(x) \quad (x \in S^1)\\
\psi(x,1) &= \phi_1(x) \quad (x \in S^1),
\end{aligned}$$

and we define *null-homotopic in* Z in the natural way. A subset $Z \subseteq \Sigma$ is *simply-connected* if (i) Z is arc-wise connected, and (ii) every continuous function $\phi\colon S^1 \to Z$ is null-homotopic in Z.

We shall frequently need the following lemma.

(3.1) *If $X, Y \subseteq \Sigma$ are closed, and X is simply-connected and $X \cap Y$ is non-empty and arc-wise connected, and every point $y \in X \cap Y$ has an open neighbourhood A such that $X \cap A$ can be deformed in X into $X \cap Y$ with the points of $X \cap Y \cap A$ kept fixed, then any continuous $\phi\colon S^1 \to X \cup Y$ is homotopic in $X \cup Y$ to a continuous $\phi'\colon S^1 \to Y$.*[F2]

[F1] **Missing proof.** The paper omits these proofs. The argument above is the one used in the formalization, which does not use the uniqueness of radial drawings.

[F2] **Statement mistake.** As printed, (3.1) is false for general closed sets. In the plane, let K be the comb $[0,1] \times \{0\} \cup \{0\} \times [0,1] \cup \bigcup_n \{1/n\} \times [0,1]$, let $X = [0,1]^2$, and let Y be K together with, for each m, a semicircle above $y = 1$ with ends $(1/(2m), 1)$ and $(1/(2m+1), 1)$. A curve that runs along the top of the square, over each semicircle, and back through X is not homotopic in $X \cup Y$ to a curve in Y, since such a curve would have to meet every semicircle and go down to $y = 0$ between any two of them, infinitely

Proof. We may assume that $|\phi| \not\subseteq X$ and $|\phi| \not\subseteq Y$. Let

$$T = \{x \in S^1 : \phi(x) \in X - Y\}.$$

Then T is open and $T \neq S^1$, and so T may be expressed as a union of disjoint open intervals, $T = \bigcup\{T_i : i \in I\}$, say. For $i \in I$, let T_i have ends $a_i, b_i \in S^1$. Then $T_i \cup \{a_i, b_i\}$ is homeomorphic to $[0,1]$. Let $\phi_i' \colon T_i \cup \{a_i, b_i\} \to X \cap Y$ be continuous with $\phi'(a_i) = \phi(a_i)$ and $\phi'(b_i) = \phi(b_i)$, and define $\phi' \colon S^1 \to Y$ by

$$\begin{aligned} \phi'(x) &= \phi_i'(x) && \text{if } x \in T_i \cup \{a_i, b_i\} \text{ for some } i \in I \\ &= \phi(x) && \text{if there is no such } i. \end{aligned}$$

Then $\phi' \colon S^1 \to Y$ is continuous. Moreover, the restriction of ϕ to $T_i \cup \{a, b\}$ is homotopic in X to ϕ_i', for each i, and so ϕ is homotopic in $X \cup Y$ to ϕ', as required. □

An O-arc F in Σ is *null-homotopic in* Z if some (or every) homeomorphism $\phi \colon S^1 \to F$ is null-homotopic in Z. A circuit C of a drawing Γ is *null-homotopic in* Z if $U(C)$ is null-homotopic in Z.

A walk in a drawing Γ is a finite sequence $v_0, e_1, v_1, e_2, \ldots, e_t, v_t$ with $t \geq 0$, where $v_0, v_1, \ldots, v_t$ are vertices of Γ, $e_1, e_2, \ldots, e_t$ are edges, and for [p. 77] $1 \leq i \leq t$ the ends of e_i are v_{i-1}, v_i. Its length is t. Its initial vertex and terminal vertex are v_0, v_t, respectively (these are its *ends*), and the walk is said to be *from* v_0 *to* v_t. Its *internal vertices* are $v_1, v_2, \ldots, v_{t-1}$. We see that if (say) $v_2 = v_0$ and $t > 2$ then v_0 is both the initial vertex and an internal vertex. The walk is *simple* if $v_0, v_1, \ldots, v_t$ are all distinct. It is *closed* if $v_0 = v_t$, and then we call v_0 its *end*. It is *simple-closed* if $v_0 = v_t$, and $v_1, v_2, \ldots, v_t$ are all distinct and $e_1, e_2, \ldots, e_t$ are all distinct. Walks of length zero are thus simple-closed. (Simple-closed walks of positive length are *not* simple, which motivates the hyphen in "simple-closed".) If W is the walk $v_0, e_1, v_1, \ldots, e_t, v_t$ then $\Gamma \mid W$ denotes the subdrawing of Γ with vertex set $\{v_0, v_1, \ldots, v_t\}$ and edge set $\{e_1, e_2, \ldots, e_t\}$. We use $U(W)$ to denote $U(\Gamma \mid W)$ for brevity. W^{-1} denotes the *reverse* of W, that is, the walk $v_t, e_t, v_{t-1}, \ldots, e_1, v_0$. If

$$\begin{aligned} W &= v_0, e_1, v_1, \ldots, e_t, v_t, \\ W' &= v_0', e_1', v_1', \ldots, e_{t'}', v_{t'}' \end{aligned}$$

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often. The proof below fails at the same point: when the intervals T_i are infinitely many, the curves ϕ_i' chosen one at a time need not fit together into a continuous ϕ'. The added hypothesis holds where the paper uses (3.1), in (6.4) and (7.8), where X is a union of closed discs bounded by circuits of a drawing. With it, (3.1) holds, by a different proof.

are walks and $v_t = v'_0$, the walk

$$v_0, e_1, v_1, \ldots, e_t, v_t, e'_1, v'_1, \ldots, e'_{t'}, v'_{t'}$$

is called the *concatenation* of W and W', and is denoted by $W + W'$. If $W+W'$ is closed then $W'+W$ is also a closed walk, and we say it is obtained from $W + W'$ by *reinitializing*.

We also wish to speak of homotopy for walks. Let $W = v_0, e_1, v_1, \ldots, e_t, v_t$ be a walk in Γ. A curve ϕ is said to *trace* W if there are numbers $0 = x_0 \leq x_1 \leq \cdots \leq x_t = 1$ such that

(i) for $0 \leq i \leq t$, $\phi(x_i) = v_i$, and

(ii) for $1 \leq i \leq t$ and $x_{i-1} \leq x \leq x_i$, $\phi(x) \in \bar{e}_i$.

Clearly for any walk W there is a curve which traces W. However, curves which trace W are not necessarily homotopic to one another, because if some e_i is a loop, our definition of "trace" does not specify the number of times that (or the sense in which, or even whether) the curve runs along e_i. However, if every loop e of W is null-homotopic in Z (that is, the O-arc $\bar{e}$ is null-homotopic in Z), it is easy to see that all curves that trace W are homotopic in Z to one another. This leads to the following definition of homotopy for walks: if W, W' are walks in Γ and $Z \subseteq \Sigma$, we say W is *homotopic in Z to W'* if

(i) every loop of W and every loop of W' is null-homotopic in Z, and

[p. 78] (ii) there exist curves ϕ, ϕ' tracing W, W', respectively, such that ϕ is homotopic in Z to ϕ'.

A walk W is *null-homotopic in Z* if it is homotopic in Z to a walk of length zero (and hence is closed).

In all the forms of homotopy mentioned above, we shall usually abbreviate "homotopic in Σ" to "homotopic."

4. Slopes

Let Σ be a surface, and let Γ be a painting or drawing in Σ. A subset $Z \subseteq \Sigma$ is Γ-*normal* if $Z \cap U(\Gamma) \subseteq V(\Gamma)$. If Σ is connected, we say that Γ is θ-*representative* if every Γ-normal O-arc F with $|F \cap V(\Gamma)| < \theta$ is null-homotopic. If $\Delta \subseteq \Sigma$ is closed and $bd(\Delta)$ is Γ-normal then $(U(\Gamma)\cap\Delta, V(\Gamma)\cap\Delta)$ is a painting or drawing (respectively) which we denote by $\Gamma \cap \Delta$. A tangle $\mathscr{T}$ of order θ in a painting or drawing Γ is said to be *respectful* if Σ is connected and for every Γ-normal O-arc F in Σ with $|F \cap V(\Gamma)| < \theta$, there is a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta) = F$ such that

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}.$$

If there is a respectful tangle of order θ then clearly Γ is θ-representative, and our first main result is the converse for non-spherical Σ, the following.

(4.1) *Let Σ be a connected surface, not a sphere, let $\theta \geq 1$, and let Γ be a θ-representative 2-cell painting or drawing in Σ. Then there is a unique respectful tangle in Γ of order θ.*

Statement (4.1) will be proved in Section 6. To prove (4.1) we use the concept of a "slope" in a drawing Γ in a connected surface Σ, defined as follows. Let $\theta \geq 1$ be half an integer. A *slope* in Γ of order θ is a function ins which assigns to every circuit C of Γ of length $< 2\theta$ a closed disc $\mathrm{ins}(C) \subseteq \Sigma$, bounded by $U(C)$, such that

(i) if C_1, C_2 are circuits of Γ, both of length $< 2\theta$, and $U(C_1) \subseteq \mathrm{ins}(C_2)$, then $\mathrm{ins}(C_1) \subseteq \mathrm{ins}(C_2)$

(ii) if P_0, P_1, P_2 are three paths of Γ of length > 0, with the same ends, mutually internally disjoint, and the circuits $P_1 \cup P_2$, $P_2 \cup P_0$, $P_0 \cup P_1$ all have length $< 2\theta$, then one of

$$\mathrm{ins}(P_1 \cup P_2), \quad \mathrm{ins}(P_2 \cup P_0), \quad \mathrm{ins}(P_0 \cup P_1)$$

includes the other two.

(Paths and circuits have no "repeated" vertices, and the length of a path or [p. 79] circuit is the number of edges in it.) We speak of these as the *first and second slope axioms*. (The factor of 2 in the definition of "order" here is for later convenience. It may be helpful to note that in our applications, Γ will be bipartite and θ will be integer.)

We see that if Γ has a slope of order θ then every circuit of Γ of length $< 2\theta$ is null-homotopic, and conversely, if Σ is connected and not a sphere and every circuit of Γ of length $< 2\theta$ is null-homotopic, then Γ has a unique slope of order θ. (This is because any null-homotopic O-arc in a surface bounds a disc—see (7.1), later.) The case when Σ is a sphere is more complicated, however. Certainly every drawing in a sphere has a slope of order θ, for any θ; because we may simply choose a region r, and define $\mathrm{ins}(C)$ to be the closed disc bounded by $U(C)$ not including r. Drawings in a sphere may also have slopes in which there is no such region r, and these turn out to be much more interesting to us.

If Γ is a θ-representative 2-cell painting or drawing in a connected non-sphere Σ, then there is clearly a slope of order θ in every radial drawing of Γ, and we shall use the properties of this slope to prove (4.1). To do so, we must first prove some general results about slopes; and it turns out that they hold even if Σ is a sphere.

Throughout this and the next section, Γ is a drawing in a connected surface Σ with a slope ins of order θ. We begin with the following, the proof of which is clear.

(4.2) *Let Γ' be a subdrawing of Γ. Then the restriction of ins to the set circuits of Γ' of length $< 2\theta$ is a slope of order θ in Γ'.*

A subdrawing Γ' of Γ is said to *capture* a subset $X \subseteq \Sigma$ if there is a circuit C of Γ' of length $< 2\theta$ with $X \subseteq \mathrm{ins}(C)$. A subdrawing Γ' of Γ is *confined* if every circuit of Γ' has length $< 2\theta$. If Γ' is confined, we define $\mathrm{ins}(\Gamma')$ to be the union of $U(\Gamma')$ and $\mathrm{ins}(C)$ for all circuits C of Γ'. We say Γ is *non-separable* if it is connected and there is no separation (A, B) of order 1 with $E(A), E(B) \neq \emptyset$.

(4.3) *Let Γ' be a confined subdrawing of Γ, and let C be a circuit of Γ' with ins(C) maximal. Then* (i) *$U(P) \subseteq \mathrm{ins}(C)$ for every path P of Γ' with both ends in $V(C)$,* (ii) *for every edge e of C, Γ' does not capture the region of Γ incident with e not included in ins(C), and* (iii) *if Γ' is non-separable then ins(Γ') = ins(C).*

Proof. Suppose that there is a path P of Γ' with $U(P) \not\subseteq \mathrm{ins}(C)$ and with both ends of P in $V(C)$. Let us choose such a path P of minimum length; then no internal vertex of P is in C. Let the ends of P be a, b, and [p. 80] let P_1, P_2 be the two paths of C between a and b. Now $P_1 \cup P_2$, $P_2 \cup P$, $P \cup P_1$ are all circuits, and all have length $< 2\theta$, since Γ' is confined. But $U(P) \not\subseteq \mathrm{ins}(P_1 \cup P_2) = \mathrm{ins}(C)$, and so by the second slope axiom, one of $\mathrm{ins}(P \cup P_1)$, $\mathrm{ins}(P \cup P_2)$ includes $\mathrm{ins}(P_1 \cup P_2) = \mathrm{ins}(C)$. This contradicts the fact that C was chosen with $\mathrm{ins}(C)$ maximal, for $\mathrm{ins}(P \cup P_1)$, $\mathrm{ins}(P \cup P_2) \neq \mathrm{ins}(C)$ since they include $U(P)$. We deduce that there is no such P, and so (i) holds. In particular, if Γ' is non-separable then $U(\Gamma') \subseteq \mathrm{ins}(C)$. By the first slope axiom, $\mathrm{ins}(\Gamma') \subseteq \mathrm{ins}(C)$. Since $\mathrm{ins}(C) \subseteq \mathrm{ins}(\Gamma')$, this proves (iii).

To complete the proof of (ii), let $e \in E(C)$, and let r be the region of Γ incident with e with $r \not\subseteq \mathrm{ins}(C)$. We suppose, for a contradiction, that Γ' captures r. Let C' be a circuit of Γ' with $r \subseteq \mathrm{ins}(C')$. If $U(C) \subseteq \mathrm{ins}(C')$, then by the first slope axiom, $\mathrm{ins}(C) \subseteq \mathrm{ins}(C')$, contrary to our choice of C. Thus $U(C) \not\subseteq \mathrm{ins}(C')$, and so there are distinct vertices $a, b \in V(C \cap C')$. But by (i), $U(P) \subseteq \mathrm{ins}(C)$ for both paths P of C' with ends a, b, and so $U(C) \subseteq \mathrm{ins}(C')$, a contradiction. We deduce that Γ' does not capture r, as claimed. □

The next result is a strengthening of (4.3).

(4.4) *If Γ' is confined then $\Sigma - \mathrm{ins}(\Gamma')$ is a region of Γ'.*

Proof. We claim first of all that there is at least one region of Γ' not captured by Γ'. For if Γ' has no circuits, then it captures no region, and

our claim is true. if Γ' has a circuit, we may choose a circuit C with $\mathrm{ins}(C)$ maximal, and our claim follows from (4.3)(ii).

It remains to show that there is at most one such region. We prove by induction on $|E(\Gamma')|$ that if r'_1, r'_2 are distinct regions of Γ' then Γ' captures at least one of them. For if Γ' has no circuits, it has only one region (since Σ is connected) and our claim is vacuous. Thus we may assume that Γ' has a circuit C. Let r' be a region of Γ' with $r' \subseteq \mathrm{ins}(C)$, and let e be an edge of Γ' incident with r'. Let Γ'' be the subdrawing of Γ' obtained by deleting e. Let r''_1, r''_2 be regions of Γ'' with $r'_1 \subseteq r''_1$, $r'_2 \subseteq r''_2$. Now Γ' captures r', and so if $r'_1 = r'$ or $r'_2 = r'$ the claim is true; we assume $r'_1, r'_2 \neq r'$. Since r' is incident with e, at least one of r'_1, r'_2 is not incident with e, say r'_1. Hence $r''_1 = r'_1$; but $r'_2 \not\subseteq r'_1$, and so $r''_1 \neq r''_2$. From our inductive hypothesis we deduce that Γ'' and hence Γ' captures at least one of r''_1, r''_2, and hence captures at least one of r'_1, r'_2, as required. □

(4.5) *Let $\Gamma' \subseteq \Gamma$ be confined, and let $r' \in R(\Gamma')$. Suppose that Γ captures r'. Then so does Γ'.*

Proof. Let C be a circuit of Γ with $r' \subseteq \mathrm{ins}(C)$. Then $\mathrm{ins}(C)$ is a closed disc, and there is a circuit C' of Γ' bounding a closed disc $\Delta' \subseteq \mathrm{ins}(C)$, with [p. 81] $r' \subseteq \Delta'$, since $r' \in R(\Gamma')$. Since Γ' is confined, C' has length $< 2\theta$; but $U(C') \subseteq \mathrm{ins}(C)$, and so by the first slope axiom, $\mathrm{ins}(C') \subseteq \mathrm{ins}(C)$, that is, $\mathrm{ins}(C') = \Delta'$. It follows that Γ' captures r', as required. □

5. Small Sets

Again, throughout this section Γ is a drawing in a connected surface Σ with a slope ins of order θ. A weighting is a pair (Ω, ω), where $\Omega \subseteq V(\Gamma)$, $\omega \geq 1$ is an integer and $|\Omega \cap V(C)| \geq \omega$ for every circuit C of Γ of length $\geq 2\theta$. For instance, $(V(\Gamma), 2\theta)$ is a weighting; or if Γ is bipartite and θ is integral, we could take Ω to be one side of a bipartition and $\omega = \theta$. Throughout this section (Ω, ω) is a fixed weighting.

Let $Z \subseteq R(\Gamma)$. We define $N_\Gamma(Z) = N(Z)$ to be the subdrawing of Γ formed by those vertices and edges of Γ incident both with a member of Z and with a member of $R(\Gamma) - Z$. We say that Z is *small* (with respect to (Ω, ω)) if $|\Omega \cap V(N(Z))| < \omega$, and $N(Z)$ captures r for each $r \in Z$. This section is devoted to proving certain properties of small sets, which will be central when we come to establish the connection between slopes and tangles.

We begin with the following lemma.

(5.1) *Let $Z \subseteq R(\Gamma)$, and let $\Gamma' \subseteq \Gamma$. Let r' be a region of Γ'. If there are*

regions $r_1 \in Z$, $r_2 \in R(\Gamma) - Z$ with $r_1, r_2 \subseteq r'$, then there are such regions r_1, r_2 which in addition are incident with some edge e of Γ with $e \notin E(\Gamma')$.

Proof. Since $r_1, r_2 \subseteq r'$ and r' is a region of Γ' there is a sequence

$$r_1 = a_1, a_2, \ldots, a_s = r_2$$

of regions $a_1, a_2, \ldots, a_s$ of Γ with $a_1, a_2, \ldots, a_s \subseteq r'$, such that for $1 \leq i < s$ there is an edge e_i of Γ incident with both a_i and a_{i+1}, not in $E(\Gamma')$. Choose i maximum such that $a_i \in Z$ (this is possible since $a_1 \in Z$); then $i \neq s$ since $a_s \notin Z$, and a_i, a_{i+1} satisfy the theorem. □

We apply (5.1) to deduce the following.

(5.2) *If $Z \subseteq R(\Gamma)$ and $|\Omega \cap V(N(Z))| < \omega$, then exactly one of Z, $R(\Gamma) - Z$ is small.*

Proof. Certainly $N(Z)$ is confined, from the definition of a weighting, and so by (4.4) there is a unique region r_0 of $N(Z)$ not captured by $N(Z)$. Thus for $r \in R(\Gamma)$, $N(Z)$ captures r if and only if $r \not\subseteq r_0$. It follows that at most one of Z, $R(\Gamma) - Z$ is small. Suppose, for a contradiction, that neither is small, so that there exist $r_1 \in Z$, $r_2 \in R(\Gamma) - Z$ with $r_1, r_2 \subseteq r_0$. By (5.1), [p. 82] we may choose r_1, r_2 both incident with some edge of Γ not in $E(N(Z))$. But this contradicts the definition of $N(Z)$, since $r_1 \in Z$, $r_2 \notin Z$, and completes the proof. □

(5.3) *If $Z \subseteq R(\Gamma)$ is small, $Z' \subseteq Z$, and $|\Omega \cap V(N(Z'))| < \omega$, then Z' is small.*

Proof. We must show that $N(Z')$ captures every $r \in Z'$. Thus, let $r \in Z'$. Then $r \in Z$, and so $N(Z)$ captures r; let C be a circuit of $N(Z)$ with $r \subseteq \mathrm{ins}(C)$. Let r' be the region of $N(Z')$ with $r \subseteq r'$. We claim that $r' \subseteq \mathrm{ins}(C)$. Suppose not. Let $X = \{s \in R(\Gamma) : s \subseteq \mathrm{ins}(C)\}$. Since $X \cap Z'$ contains some but not every region of Γ included in r', by (5.1) there are regions r_1, r_2 of Γ with $r_1, r_2 \subseteq r'$, and with $r_1 \in X \cap Z'$, $r_2 \notin X \cap Z'$, both incident with some edge e of Γ not in $N(Z')$. Since $e \notin E(N(Z'))$ and $r_1 \in Z'$ it follows that $r_2 \in Z'$, and so $r_2 \notin X$. Since $r_1 \in X$ we deduce that $e \in E(C) \subseteq E(N(Z))$. But this is impossible since $r_1, r_2 \in Z' \subseteq Z$. This proves our claim that $r' \subseteq \mathrm{ins}(C)$, and so Γ captures r'. By (4.5), $N(Z')$ captures r' and hence r, as required. □

The main and most difficult result of this section is the following. (We say (Z_1, Z_2, Z_3) is a partition of Z if Z_1, Z_2, Z_3 are disjoint sets whose union is Z.)

(5.4) *If (Z_1, Z_2, Z_3) is a partition of $R(\Gamma)$, then one of Z_1, Z_2, Z_3 is not small.*

Proof. We denote $N(Z_1) \cup N(Z_2) \cup N(Z_3)$ by $N(Z_1, Z_2, Z_3)$. If possible, let (Z_1, Z_2, Z_3) be a counterexample chosen with $N(Z_1, Z_2, Z_3)$ minimal. Then Z_1, Z_2, Z_3 are small. Therefore for $i = 1, 2, 3$ we have $|\Omega \cap V(C)| < \omega$ and consequently $|V(C)| < 2\theta$ for every circuit C in $N(Z_i)$, from which it follows that $N(Z_i)$ is confined and $\mathrm{ins}(C)$ is defined for every circuit C in $N(Z_i)$. Moreover $Z_1 \neq \emptyset$ because otherwise a contradiction would be obtained by taking $Z = Z_2$ in (5.2), and similarly Z_2, Z_3 are non-empty.

A Z_1-ring is a circuit C of $N(Z_1)$ chosen with $\mathrm{ins}(C)$ maximal. If C_1 is a Z_1-ring and C is a circuit of $N(Z_1)$ with $|V(C) \cap \mathrm{ins}(C_1)| \geq 2$, then $U(C) \subseteq \mathrm{ins}(C_1)$ by (4.3)(i). Consequently, if C_1, C_2 are distinct Z_1-rings and C is a circuit of $N(Z_1)$ with $|V(C \cap C_i)| \geq 2$ $(i = 1, 2)$, then $|V(C_2) \cap \mathrm{ins}(C_1)| \geq 2$ since $U(C) \subseteq \mathrm{ins}(C_1)$; and so $U(C_2) \subseteq \mathrm{ins}(C_1)$, and consequently $\mathrm{ins}(C_2) \subseteq \mathrm{ins}(C_1)$ by the first slope axiom. But similarly $\mathrm{ins}(C_1) \subseteq \mathrm{ins}(C_2)$, and so equality holds, which is impossible since $C_1 \neq C_2$. Thus there do not exist such C_1, C_2, C. In particular, any two distinct Z_1-rings intersect in at most one vertex and hence have no common edge, and also if e, f are edges of distinct Z_1-rings then no circuit of $N(Z_1)$ [p. 83] contains them both. Let $\mathscr{R}$ be the set of all Z_1-rings, and $\mathscr{S}$ the set of all vertices of Γ in at least two Z_1-rings. Let G be the graph with vertex set $\mathscr{R} \cup \mathscr{S}$, in which $C \in \mathscr{R}$ is adjacent to $v \in \mathscr{S}$ if $v \in V(C)$. It follows that G is a forest. Moreover G is non-empty since Z_1, Z_2, Z_3 are non-empty. Since every vertex of G in $\mathscr{S}$ has valency ≥ 2 in G, we deduce that some vertex of G in $\mathscr{R}$ has valency ≤ 1; that is (since every vertex of $N(Z_1)$ has even valency and so every edge of $N(Z_1)$ is in a circuit),

(1) *There is a Z_1-ring C_1 such that at most one vertex of C_1 is incident with an edge of $N(Z_1)$ not in* $\mathrm{ins}(C_1)$.

Let C_1 be as in (1). Choose $v_1 \in V(C_1)$ such that no $v \in V(C_1)$ with $v \neq v_1$ is incident with an edge of $N(Z_1)$ not in $\mathrm{ins}(C_1)$. Let f be an edge of C_1 incident with v_1. Let $A = \{r \in R(\Gamma) : r \subseteq \mathrm{ins}(C_1)\}$ and $B = R(\Gamma) - A$. For each $v \in V(C_1)$ let $R(v)$ be the set of those members of B which are incident with v.

If $e \in E(C_1)$ then e is incident with just two regions of Γ, regions $r \in A$ and $s \in B$ say. In these circumstances, $N(Z_1)$ does not capture s by (4.3)(ii) and so, since Z_1 is small, $s \notin Z_1$. Therefore, since $e \in E(C_1) \subseteq E(N(Z_1))$, we must have $r \in Z_1$. We conclude that

(2) *If $e \in E(C_1)$ then one region of Γ incident with e belongs to $A \cap Z_1$ and the other belongs to $B \cap (Z_2 \cup Z_3)$.*

It follows from (2) that $R(v) \subseteq Z_2 \cup Z_3$ for all $v \neq v_1$. For $i = 2, 3$, let

$$W_i = \{v \in V(C_1) : v \neq v_1, R(v) \subseteq Z_i\}.$$

From the symmetry, we may assume that $|W_2 \cap \Omega| \geq |W_3 \cap \Omega|$, and if equality holds then $f \in E(N(Z_2))$.

(3) $E(C_1 \cap N(Z_2)) \neq \emptyset$.

Subproof. If $W_2 \neq \emptyset$ this is clear. If $W_2 = \emptyset$ then since $|W_2 \cap \Omega| \geq |W_3 \cap \Omega|$, we deduce that equality holds and so $f \in E(N(Z_2))$, as required.

For $i = 1, 2, 3$ let $Z_i' = Z_i \cap B$.

(4) *For $i = 1, 2, 3$, $N(Z_i') \subseteq N(Z_i)$, and Z_i' is small.*

Subproof. Let $e \in N(Z_i')$, incident with regions r, s, where $r \in Z_i'$ and $s \notin Z_i'$. Then $r \in Z_i$, and $r \not\subseteq \text{ins}(C_1)$. We claim that $s \notin Z_i$. For if $s \in B$ then $s \notin Z_i$ since $s \notin Z_i'$. If $s \in A$ then $e \in E(C_1)$, and by (2) $r \notin Z_1$ and $s \in Z_1$. Consequently $i \neq 1$ (since $r \in Z_i$) and $s \notin Z_i$ (since $s \in Z_1$). Thus in either [p. 84] case $s \notin Z_i$, as claimed. We deduce that $e \in N(Z_i)$. The first claim follows. Since

$$|V(N(Z_i')) \cap \Omega| \leq |V(N(Z_i)) \cap \Omega| < \omega$$

and $Z_i' \subseteq Z_i$, the second claim follows from (5.3).

(5) $|V(N(Z_2 \cup A)) \cap \Omega| \leq |V(N(Z_2)) \cap \Omega|$.

Subproof. Let $V(N(Z_2)) = X$, $V(N(Z_2 \cup A)) = Y$, $\{v_1\} - V(N(Z_2)) = D$. (Thus $|D| \leq 1$.) Then $W_2 \subseteq X - Y$, and $Y - X \subseteq W_3 \cup D$, for $N(Z_2 \cup A) \subseteq N(Z_2) \cup N(A) = N(Z_2) \cup C_1$. Hence

$$\begin{aligned} |X \cap \Omega| - |Y \cap \Omega| &= |(X - Y) \cap \Omega| - |(Y - X) \cap \Omega| \\ &\geq |W_2 \cap \Omega| - |W_3 \cap \Omega| - |D|. \end{aligned}$$

But $|W_2 \cap \Omega| \geq |W_3 \cap \Omega|$, and if equality occurs then $f \in E(N(Z_2))$ and therefore $|D| = 0$. Hence $|X \cap \Omega| \geq |Y \cap \Omega|$ as required.

Let $K = N(Z_1, Z_2, Z_3)$.

(6) *$E(N(Z_1) \cap N(Z_3)) \subseteq E(C_1)$, and $\text{ins}(C_1) - U(C_1)$ is a region of K, and $A \subseteq Z_1$.*

Subproof. $(Z_1', Z_2 \cup A, Z_3')$ is a partition of $\mathscr{R}(\Gamma)$, and Z_1', Z_3' are small by (4). Moreover, $N(Z_1', Z_2 \cup A, Z_3')$ is a subgraph of K and it is a proper subgraph because some edge of C_1 belongs to $N(Z_2)$ by (3), and hence, by (2), does not belong to $N(Z_1', Z_2 \cup A, Z_3')$. From the choice of Z_1, Z_2, Z_3 it follows that $Z_2 \cup A$ is not small. Moreover $|\Omega \cap V(N(Z_2 \cup A))| < \omega$ by (5). From (5.2) we deduce that $\mathscr{R}(\Gamma) - (Z_2 \cup A) = Z_1' \cup Z_3'$ is small. Hence $(A, Z_2', Z_1' \cup Z_3')$ is a partition of $\mathscr{R}(\Gamma)$ into three small sets, by (4) (for A is small since $N(A) = C_1$). Moreover, $N(A, Z_2', Z_1' \cup Z_3') \subseteq K$, and so equality holds, from the choice of Z_1, Z_2, Z_3. Let $e \in E(N(Z_1) \cap N(Z_3))$. Then e is incident with no region of Z_2 and hence with none of Z_2'; and since $e \in E(N(A, Z_2', Z_1' \cup Z_3'))$ it follows that $e \in E(N(A)) = E(C_1)$. This

proves the first claim. The second is clear, for $\mathrm{ins}(C_1) - U(C_1)$ includes no edges of $N(A, Z_2', Z_1' \cup Z_3')$. The third follows from (5.1), for $\mathrm{ins}(C_1) - U(C_1)$ certainly includes at least one region in Z_1.

(7) $Z_1 = A$.

Subproof. Suppose that $Z_1 \neq A$. By (6), there is another Z_1-ring, and so the graph G defined before (1) has at least two vertices of valency ≤ 1. Let $C_1' \neq C_1$ be another Z_1-ring satisfying (1). By (6) applied to C_1', there exists $j \in \{2, 3\}$ such that $E(N(Z_1) \cap N(Z_j)) \subseteq E(C_1')$. But $j \neq 2$, by (3), since [p. 85] $E(C_1 \cap C_1') = \emptyset$, and so $j = 3$. By (6), $E(N(Z_1) \cap N(Z_3)) = \emptyset$. Consequently, $N(Z_1), N(Z_3) \subseteq N(Z_2)$ and so $N(Z_2)$ captures every region of Γ, contrary to (4.4). This contradiction shows that $Z_1 = A$.

(8) *For $i = 1, 2, 3$, $N(Z_i)$ is a circuit C_i say, and $Z_i = \{r \in R(\Gamma) : r \subseteq ins(C_i)\}$, and $ins(C_i) - U(C_i)$ is a region of K. In particular, K has precisely three regions.*

Subproof. By (7), $Z_1 = \{r \in R(\Gamma) : r \subseteq \mathrm{ins}(C_1)\}$ and therefore $N(Z_1) = C_1$ and $\mathrm{ins}(C_1) - U(C_1)$ is a region of K; and similar arguments apply to Z_2, Z_3. Since $r_i = \mathrm{ins}(C_i) - U(C_i)$ includes all members of Z_i and $Z_1 \cup Z_2 \cup Z_3 = R(\Gamma)$, it follows that K has no regions other than r_1, r_2, r_3.

(9) *Every circuit of K is null-homotopic in Σ.*

Subproof. Every circuit C of K of length $< 2\theta$ is null-homotopic in $\mathrm{ins}(C)$ and hence in Σ. In particular, C_1 and C_2 are null-homotopic in Σ. Since $\theta \geq 1$, every loop of K is null-homotopic, and so we may speak of homotopy for walks. Now $C_1 \cup C_2 = K$; let W be a simple-closed walk of K, and let

$$V(\Gamma|W) \cap V(C_1) \cap V(C_2) = \{v_1, \ldots, v_t\}$$

numbered in order on W. We shall show that W is null-homotopic. If $t \leq 1$ then either $\Gamma|W = C_1$ or $\Gamma|W = C_2$ and the result is true. We assume then that $t \geq 2$. Let $v_0 = v_t$, and for $1 \leq i \leq t$ let P_i be a simple walk of $\Gamma|W$ from v_{i-1} to v_i, with no internal vertex in $V(C_1) \cap V(C_2)$, such that $P_1 + P_2 + \cdots + P_t$ is a reinitializing of W. For $1 \leq i \leq t$, let Q_i be a simple walk of C_1 from v_{i-1} to v_i. If for all i with $1 \leq i \leq t$ P_i is homotopic to Q_i, then some reinitializing of W is homotopic to a closed walk of C_1, and hence W is null-homotopic, as required. We assume, for a contradiction, that P_1 is not homotopic to Q_1 (say). Now P_1 is a simple walk of $C_1 \cup C_2$ with no internal vertex in $V(C_1 \cap C_2)$, and so P_1 is a walk of one of C_1, C_2. It is not a walk of C_1, since it is not homotopic to Q_1, and so P_1 is a walk of C_2. Let P_1' be the other simple walk of C_2 from v_0 to v_1 and let Q_1' be the simple walk of C_1 different from Q_1 from v_0 to v_1. Then P_1 is homotopic to P_1', and Q_1 is homotopic to Q_1', and so neither of P_1, P_1' is homotopic to

either of Q_1, Q_1'. Now some one, say P_1'', of P_1, P_1' has length $< \theta$, since C_2 has length $< 2\theta$, and similarly some one, say Q_1'', of Q_1, Q_1' has length $< \theta$. Then $P_1'' + Q_1''^{-1}$ has length $< 2\theta$, and is not null-homotopic, which is impossible. Thus W is null-homotopic. The result follows.

Now by (8), the closure of every region of K is a closed disc, and so by (9) and (3.1), Σ is simply-connected. Hence Σ is a sphere. From (8), K is [p. 86] a planar graph with three regions, each bounded by a circuit; and so $C_1 \cap C_2$, $C_1 \cap C_3$, $C_2 \cap C_3$ are all paths joining the same pair of vertices. This contradicts the second slope axiom. Hence our choice of Z_1, Z_2, Z_3 was impossible, and the result follows. □

The following is a slight extension of (5.4).

(5.5) *If $Z_1, Z_2, Z_3 \subseteq R(\Gamma)$ are small, then $Z_1 \cup Z_2 \cup Z_3 \neq R(\Gamma)$.*

Proof. We suppose, for a contradiction, that the result is false. Let us choose Z_1, Z_2, $Z_3 \subseteq R(\Gamma)$, small, with union $R(\Gamma)$, and with $|Z_1|+|Z_2|+|Z_3|$ minimum. Let $Z_1' = Z_1 - Z_2$, $Z_2' = Z_2 - Z_1$. We claim that

$$V(N(Z_1')) \cup V(N(Z_2')) \subseteq V(N(Z_1)) \cup V(N(Z_2)).$$

For suppose that some vertex v is contained in, say, $V(N(Z_1'))$. Let r_1, r_2 be regions incident with v, such that $r_1 \in Z_1'$, $r_2 \notin Z_1'$. If $r_2 \notin Z_1$, then $v \in V(N(Z_1))$, and so we may assume that $r_2 \in Z_1$. Since $r_2 \notin Z_1'$, it follows that $r_2 \in Z_2$. But $r_1 \notin Z_2$, and so $v \in V(N(Z_2))$. This proves our claim.

Next we claim

$$V(N(Z_1')) \cap V(N(Z_2')) \subseteq V(N(Z_1)) \cap V(N(Z_2)).$$

For suppose that $v \in V(N(Z_1')) \cap V(N(Z_2'))$. Then certainly there are regions $r_1 \in Z_1'$, $r_2 \in Z_2'$ incident with v. By definition of Z_1', Z_2', we see that $r_1 \notin Z_2$, $r_2 \notin Z_1$; and so $v \in V(N(Z_1)) \cap V(N(Z_2))$. This proves our claim.

From these two facts, we deduce that

$$\begin{aligned}
&|\Omega \cap V(N(Z_1'))| + |\Omega \cap V(N(Z_2'))| \\
&\quad = |\Omega \cap (V(N(Z_1')) \cup V(N(Z_2')))| + |\Omega \cap (V(N(Z_1')) \cap V(N(Z_2')))| \\
&\quad \leq |\Omega \cap (V(N(Z_1)) \cup V(N(Z_2)))| + |\Omega \cap (V(N(Z_1)) \cap V(N(Z_2)))| \\
&\quad = |\Omega \cap V(N(Z_1))| + |\Omega \cap V(N(Z_2))| \leq 2\omega - 2.
\end{aligned}$$

Without loss of generality, then, we may assume that $|\Omega \cap V(N(Z_1'))| \leq \omega - 1$, and so Z_1' is small, by (5.3). But $Z_1' \cup Z_2 \cup Z_3 = R(\Gamma)$, and hence by our choice of Z_1, Z_2, Z_3, we deduce that $Z_1' = Z_1$, that is, $Z_1 \cap Z_2 = \emptyset$. Similarly

$Z_1 \cap Z_3 = Z_2 \cap Z_3 = \emptyset$ and so (Z_1, Z_2, Z_3) is a partition of $R(\Gamma)$, which is impossible, by (5.4). This completes the proof of (5.5). □

Incidentally, our device of "weightings" in this section seems artificial, and in view of (5.5) and its application in the next section, one might hope that something more general and more natural was true. Let us say $Z \subseteq R(\Gamma)$ is *little* if $N(Z)$ is confined, and $N(Z)$ captures each member of Z. One [p. 87] might hope that no three little sets can have union $R(\Gamma)$; but that is false. For example, let Γ be a 2θ-representative cubic drawing in a torus, such that each region is bounded by a circuit of length $< 2\theta$ and Γ is 3-face-colourable. Let ins be the (unique) slope in Γ of order θ. Since Γ is 3-face-colourable there is a partition (Z_1, Z_2, Z_3) of $R(\Gamma)$ such that for $1 \le i \le 3$, no two regions in Z_i have a common edge. Since Γ is cubic it follows that no two regions in Z_i have a common vertex, and so $N(Z_i)$ is confined. Hence each Z_i is little, and yet $Z_1 \cup Z_2 \cup Z_3 = R(\Gamma)$.

6. Tangles And Slopes

Now we shall prove (4.1). To do so, we need to clarify the connection between tangles and slopes. Let us say a pretangle of order $\theta \ge 1$ in a hypergraph G is a set $\mathscr{T}$ of separations of G, all of order $< \theta$, satisfying the first and second tangle axioms. The definition of "respectful" for pretangles is the same as for tangles.

Let Γ be a 2-cell painting or drawing in a connected surface Σ, and let Γ' be a radial drawing of Γ. Let $\mathscr{T}$ be a respectful pretangle in Γ of order θ. For each circuit C of Γ' of length $< 2\theta$, $U(C)$ is a Γ-normal O-arc with $|U(C) \cap V(\Gamma)| < \theta$, and so there is a (unique) closed disc $\Delta \subseteq \Sigma$ with $\mathrm{bd}(\Delta) = U(C)$ and with

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}.$$

We define $\mathrm{ins}(C) = \Delta$. Then ins is a slope in Γ' of order θ, as is easily seen. We call it the slope derived from $\mathscr{T}$.

Conversely, we have the following.

(6.1) *Let Γ be a 2-cell painting or drawing in a connected surface Σ, with $E(\Gamma) \neq \emptyset$, and let Γ' be a radial drawing of Γ. For $A \subseteq \Gamma$, let $Z(A)$ denote the set of regions of Γ' which include an edge of A. Let $\theta \ge 1$ be an integer, let ins be a slope in Γ' of order θ, and let $\mathscr{T}$ be the set of all separations (A, B) of Γ of order $< \theta$ such that $Z(A)$ is small with respect to the weighting $(V(\Gamma), \theta)$ of Γ'. Then $\mathscr{T}$ is a respectful pretangle in Γ and ins is the slope derived from it.*

Proof. First, let us verify that $\mathscr{T}$ satisfies the first two tangle axioms.

Let (A, B) be a separation of Γ of order $< \theta$. Now $Z(A) = R(\Gamma') - Z(B)$ (for every region of Γ' includes a unique edge of Γ, since $E(\Gamma) \neq \emptyset$), and so $N(Z(A)) = N(Z(B))$. Thus $V(N(Z(A))) \cap V(\Gamma) \subseteq V(A \cap B)$ and so $|V(N(Z(A))) \cap V(\Gamma)| < \theta$, and therefore $N(Z(A))$ is confined. By (5.2), exactly one of $Z(A), Z(B)$ is small, and the first tangle axiom is satisfied. The second follows from (5.5). Thus $\mathscr{T}$ is a pretangle of order θ.

[p. 88] Let us verify that $\mathscr{T}$ is respectful. Let F be a Γ-normal O-arc with $|F \cap V(\Gamma)| < \theta$. If $|F \cap V(\Gamma)| = 0$ then $F \subseteq r$ for some region r of Γ, and since Γ is 2-cell there is a closed disc $\Delta \subseteq r$ with $F = bd(\Delta)$. Since $\emptyset = Z(\Gamma \cap \Delta)$ is small it follows that $(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$, as required. Thus we may assume that $|F \cap V(\Gamma)| \geq 1$. We proceed by induction on $|F \cap V(\Gamma)| = k$ say. Suppose first that $F \cap r$ is connected for every region r of Γ such that $F \cap r$ is non-null. Then $F \cap r \neq \emptyset$ for precisely k regions r of Γ, and we may assume, by perturbing F slightly, that $F = U(C)$ for a circuit C of Γ' (unless $k = 1$ and F is null-homotopic, when we are finished). Since $|F \cap V(\Gamma)| < \theta$, it follows that C has length $< 2\theta$, and so $ins(C)$ exists; let $\Delta = ins(C)$, and let $A = \Gamma \cap \Delta$, $B = \Gamma \cap \overline{\Sigma - \Delta}$. Then $Z(A)$ is small, because $C = N(Z(A))$, and so $(A, B) \in \mathscr{T}$ as required. Thus we may assume that for some region r of Γ, $F \cap r$ has more than one component.

Choose a minimal line $F' \subseteq r$ with ends in different components of $F \cap r$, and let F_1, F_2, F be the three O-arcs included in $F \cup F'$. Then F_i is Γ-normal and $|F_i \cap V(\Gamma)| < k$ $(i = 1, 2)$, and so from the inductive hypothesis, there are closed discs Δ_i such that $bd(\Delta_i) = F_i$ and

$$(\Gamma \cap \Delta_i, \Gamma \cap \overline{\Sigma - \Delta_i}) \in \mathscr{T}$$

for $i = 1, 2$. If $\Delta_1 \subseteq \Delta_2$ then since $F_1 \cap F_2 = F'$, it follows that $\Delta = \overline{\Delta_2 - \Delta_1}$ is a closed disc and $bd(\Delta) = F$. But then $\Gamma \cap \Delta \subseteq \Gamma \cap \Delta_2$ and $(\Gamma \cap \Delta_2, \Gamma \cap \overline{\Sigma - \Delta_2}) \in \mathscr{T}$, and so $(\Gamma \cap \overline{\Sigma - \Delta}, \Gamma \cap \Delta) \notin \mathscr{T}$ since $(\Gamma \cap \Delta_2) \cup (\Gamma \cap \overline{\Sigma - \Delta}) = \Gamma$. Consequently,

$$(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T},$$

as required. Thus we may assume that $\Delta_1 \not\subseteq \Delta_2$ and similarly $\Delta_2 \not\subseteq \Delta_1$. Hence $\Delta_1 \cap \Delta_2 = F'$, and $\Delta = \Delta_1 \cup \Delta_2$ is a closed disc with $bd(\Delta) = F$. Moreover,

$$(\Gamma \cap \Delta_1) \cup (\Gamma \cap \Delta_2) \cup (\Gamma \cap \overline{\Sigma - \Delta}) = \Gamma$$

and so $(\Gamma \cap \overline{\Sigma - \Delta}, \Gamma \cap \Delta) \notin \mathscr{T}$, by the second tangle axiom: and hence $(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$ as required. This proves that $\mathscr{T}$ is respectful.

It remains to check that ins is the slope derived from $\mathscr{T}$. To see this, let C be a circuit of Γ' of length $< 2\theta$. We must verify that $(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in$

$\mathscr{T}$, where $\Delta = ins(C)$. Certainly this separation has order $< \theta$, and so it suffices to show that $Z(\Gamma \cap \Delta)$ is small. But every element of $Z(\Gamma \cap \Delta)$ is a subset of $ins(C)$, and $|V(C) \cap V(\Gamma)| < \theta$, and so $Z(\Gamma \cap \Delta)$ is small, as required. □

We call $\mathscr{T}$ defined as in (6.1) the *pretangle induced from ins*. Next, we wish to show that this provides a 1-1 correspondence between respectful [p. 89] pretangles in Γ and slopes in Γ'; in other words, that if ins is derived from $\mathscr{T}$ then $\mathscr{T}$ is induced from ins. To prove that we use the following lemma.

(6.2) *Let $\mathscr{T}$ be a pretangle in a hypergraph G, and let $(A, B) \in \mathscr{T}$. Then there do not exist $(A_1, B_2), \ldots, (A_k, B_k) \in \mathscr{T}$ such that $V(A_i \cap B_i) \subseteq V(A \cap B)$ $(1 \le i \le k)$ and $B \subseteq A_1 \cup \cdots \cup A_k$.*

Proof. Suppose that such $(A_1, B_1), \ldots, (A_k, B_k)$ exist. For $0 \le i \le k$, let $A'_i = A \cup A_1 \cup \cdots \cup A_i$, $B'_i = B \cap B_1 \cap \cdots \cap B_i$. Thus $(A'_0, B'_0) = (A, B) \in \mathscr{T}$, and we prove by induction on i that $(A'_i, B'_i) \in \mathscr{T}$ for $0 \le i \le k$. Suppose then that $1 \le i \le k$, and $(A'_{i-1}, B'_{i-1}) \in \mathscr{T}$. Now $A'_i = A'_{i-1} \cup A_i$, $B'_i = B'_{i-1} \cap B_i$, and so $(B'_i, A'_i) \notin \mathscr{T}$ by the second tangle axiom, since $A_i \cup A'_{i-1} \cup B'_i = G$. But $A'_i \cap B'_i \subseteq A \cap B$ and so (A'_i, B'_i) has order $\le |V(A \cap B)|$; and hence $(A'_i, B'_i) \in \mathscr{T}$, by the first axiom. This completes the inductive argument, and we deduce that $(A'_k, B'_k) \in \mathscr{T}$. But $A'_k = A \cup B = G$, contrary to the second tangle axiom. □

We deduce

(6.3) *Let Γ be a 2-cell painting or drawing in a connected surface Σ with $E(\Gamma) \ne \emptyset$, and let $\mathscr{T}$ be a respectful pretangle in Γ of order $\theta \ge 1$. Let Γ' be a radial drawing of Γ, and let ins be the derived slope. Then for every separation (A, B) of Γ of order $< \theta$, the following are equivalent:*

(i) $(A, B) \in \mathscr{T}$

(ii) *there are circuits $C_1, \ldots, C_k$ of Γ' with $V(C_i) \cap V(\Gamma) \subseteq V(A \cap B)$ $(1 \le i \le k)$, such that every edge of A is a subset of one of $ins(C_1), \ldots, ins(C_k)$.*

In particular, $\mathscr{T}$ is the pretangle induced from ins.

Proof. We observe first that (ii) ⇒ (i). For suppose that $(A, B), C_1, \ldots, C_k$ satisfy (ii). Let $\Delta_i = ins(C_i)$ and let $A_i = \Gamma \cap \Delta_i$, $B_i = \Gamma \cap \overline{\Sigma - \Delta_i}$; then $(A_i, B_i) \in \mathscr{T}$ $(1 \le i \le k)$. Let $A_0 = A \cap B$, $B_0 = \Gamma$; then $(A_0, B_0) \in \mathscr{T}$ by the first axiom, since $(B_0, A_0) \notin \mathscr{T}$ by the second. Moreover, $A \subseteq A_0 \cup A_1 \cup \cdots \cup A_k$; for any edge of A is an edge of one of $A_1, \ldots, A_k$, by hypothesis, and any isolated vertex of A belongs to B (since Γ has no isolated vertices) and hence to A_0. Since $A_i \cap B_i \subseteq A \cap B$ for $0 \le i \le k$, we deduce from (6.2) that $(B, A) \notin \mathscr{T}$, and so $(A, B) \in \mathscr{T}$ by the first axiom. This proves (ii) ⇒ (i).

Now we prove (i) $\Rightarrow$ (ii). Let $\mathscr{T}'$ be the pretangle induced from *ins*. By definition of $\mathscr{T}'$, every member (A,B) of $\mathscr{T}'$ satisfies (ii). Let $(A,B) \in \mathscr{T}$; then $(B,A) \notin \mathscr{T}$, and so (since (ii) $\Rightarrow$ (i)) (B,A) does not satisfy (ii) (with A,B exchanged). Hence $(B,A) \notin \mathscr{T}'$, and so $(A,B) \in \mathscr{T}'$; and hence (A,B) [p. 90] satisfies (ii). This argument also shows that $\mathscr{T} \subseteq \mathscr{T}'$, and hence $\mathscr{T} = \mathscr{T}'$, as required. □

Here is a convenient way to recognize whether $(A,B) \in \mathscr{T}$ or $(B,A) \in \mathscr{T}$ when Σ is not a sphere. (This proof uses two topological results, which it is convenient to postpone to the next section.)

(6.4) *Let Γ be a 2-cell painting or drawing in a connected surface Σ which is not a sphere, and let $\mathscr{T}$ be a respectful pretangle in Γ, of order $\theta \geq 1$. Let (A,B) be a separation of Γ of order $< \theta$. Then the following are equivalent:*

(i) $(A,B) \in \mathscr{T}$

(ii) *$U(A)$ includes no non-null-homotopic O-arc*

(iii) *$U(B) - U(A)$ includes a non-null-homotopic O-arc.*

In particular, $\mathscr{T}$ is a tangle.

Proof. Since Γ is 2-cell and Σ is not a sphere, it follows that $E(\Gamma) \neq \emptyset$. Let Γ' be a radial drawing of Γ, and *ins* the derived slope.

(1) *If $(A,B) \in \mathscr{T}$ then $U(A)$ includes no non-null-homotopic O-arc.*

Subproof. Let $C_1, \ldots, C_k$ be as in (6.3). We may assume that $ins(C_i) \not\subseteq ins(C_j)$ for $i \neq j$, by choosing k minimum, and hence $U(C_1) \cup \cdots \cup U(C_k)$ is obtained from $ins(C_1) \cup \cdots \cup ins(C_k)$ by deleting finitely many disjoint open discs. If $U(A)$ includes a non-null-homotopic O-arc then so does $ins(C_1) \cup \cdots \cup ins(C_k)$, and it follows (by (3.1)) that there is a non-null-homotopic continuous function $\phi\colon S^1 \to \Sigma$ with $|\phi| \subseteq U(C_1) \cup \cdots \cup U(C_k)$. By (7.3) below there is a non-null-homotopic O-arc included in $U(C_1) \cup \cdots \cup U(C_k)$, that is, there is a non-null-homotopic circuit C of $C_1 \cup \cdots \cup C_k$. But $V(C_1 \cup \cdots \cup C_k) \cap V(\Gamma) \subseteq V(A \cap B)$, and so $|V(C) \cap V(\Gamma)| < \theta$, and hence C has length $< 2\theta$, a contradiction. This proves (1).

(2) *If $U(A)$ includes no null-null-homotopic O-arc, then $U(B) - U(A)$ includes a non-homotopic O-arc.*

Subproof. Clearly Γ is θ-representative. If Γ is a painting, for each edge $e \in E(\Gamma)$, choose a closed disc $p(e) \subseteq \bar{e}$ with $p(e) \cap V(\Gamma) = \bar{e} \cap (V(\Gamma) - V(A \cap B))$; then

$$\left(\bigcup (p(e)\colon e \in E(\Gamma)), V(\Gamma) - V(A \cap B)\right) = \Gamma^*$$

is a painting in Σ, and it is easy to see that Γ^* is $(\theta - |V(A \cap B)|)$-repre- [p. 91] sentative. (Remove the vertices one at a time.) In particular, Γ^* is 1-representative, and so there is no closed disc $\Delta \subseteq \Sigma$ with $U(\Gamma^*) \subseteq \Delta$, for

otherwise $\Sigma - \Delta$ would be a subset of a region of Γ^* and yet would include a non-null-homotopic O-arc, since Σ is not a sphere. By (7.5) below, there is a non-null-homotopic O-arc $F \subseteq U(\Gamma^*) \subseteq U(\Gamma) - V(A \cap B)$. Since (A, B) is a separation, it follows that $F \subseteq U(A) - U(B)$ or $F \subseteq U(B) - U(A)$, and the latter must hold, since $U(A)$ includes no non-null-homotopic O-arc. This proves (2) if Γ is a painting. On the other hand, if Γ is a drawing, then $\Gamma \setminus V(A \cap B)$ is 1-representative, and a similar argument (via (7.5)) applies.

(3) *If* $U(B) - U(A)$ *includes a non-null-homotopic* O*-arc then* $(A, B) \in \mathscr{T}$.

Subproof. $(B, A) \notin \mathscr{T}$ by (1), and so $(A, B) \in \mathscr{T}$ by the first tangle axiom.

This proves the equivalence of (i), (ii), and (iii).

For each $e \in E(\Gamma)$, let $K_e \subseteq \Gamma$ be the painting or drawing formed by e and its ends. We claim that if e has $< \theta$ ends then $U(K_e)$ includes no non-null-homotopic O-arc. This is clear if Γ is a painting, and so we assume that Γ is a drawing; and again the claim is true unless e is a loop of Γ such that $\bar{e}$ is a non-null-homotopic O-arc. But then Γ is not 2-representative, and since $\mathscr{T}$ is respectful of order θ, it follows that $\theta = 1$, contradicting that e has $< \theta$ ends. This proves that $U(K_e)$ includes no non-null-homotopic O-arc. Since $(K_e, \Gamma \setminus e)$ has order equal to the number of ends of e, and hence has order $< \theta$, we deduce from the equivalence of (i) and (ii) that $(K_e, \Gamma \setminus e) \in \mathscr{T}$. Therefore, $\mathscr{T}$ is a tangle, by [4, Theorem (2.7)], as required. □

Now we can prove (4.1).

Proof of (4.1). Let Γ' be a radial drawing of Γ. Certainly $E(\Gamma) \neq \emptyset$ since Γ is 2-cell and Σ is not a sphere. Since Γ is θ-representative, every circuit of Γ' of length $< 2\theta$ is null-homotopic; and this defines a slope *ins* in Γ' of order θ. Let $\mathscr{T}$ be the induced pretangle; then by (6.1), $\mathscr{T}$ is respectful and by (6.4), $\mathscr{T}$ is a tangle. $\mathscr{T}$ is unique by (6.3), since *ins* is the unique slope in Γ' of order θ. □

If r is a region of a drawing Γ, the *perimeter* of r is the drawing $\Gamma \cap \bar{r}$; and its *length* is $\sum_{e \in E(\Gamma \cap \bar{r})} n(e)$, where $n(e) = 1$ or 2, depending whether e is incident with another region or not. A slope *ins* in Γ of order θ is *even* if for every region r of Γ with perimeter of length $< 2\theta$, there is a circuit C of Γ of length $< 2\theta$ with $r \subseteq ins(C)$.

(6.5) *Let* Γ *be a 2-cell painting or drawing in a connected surface* Σ *with* [p. 92] $E(\Gamma) \neq \emptyset$, *and let* $\mathscr{T}$ *be a respectful pretangle in* Γ. *Let* Γ' *be a radial drawing of* Γ, *and let* ins *be the slope in* Γ' *derived from* $\mathscr{T}$. *Then* $\mathscr{T}$ *is a tangle if and only if* ins *is even. The "if" part assumes that* $\mathscr{T}$ *has order*

$\neq 2$ or that Γ has no loop.[F3]

Proof. Let $\mathscr{T}$ have order $\theta \geq 1$. Suppose that $\mathscr{T}$ is a tangle and let $r \in R(\Gamma')$ with perimeter of length $< 2\theta$. Let $e \in E(\Gamma)$ with $e \subseteq r$; then $V(\Gamma) \cap \bar{r}$ is the set of ends of e, and the perimeter of r has length at least twice the number of ends of e, and so e has $< \theta$ ends. Let K_e be the hypergraph formed by e and its ends; then $(\Gamma \setminus e, K_e) \notin \mathscr{T}$ by the third tangle axiom, and so $(K_e, \Gamma \setminus e) \in \mathscr{T}$ by the first. By (6.3) there is a circuit C of Γ' with

$$V(C) \cap V(\Gamma) \subseteq V(K_e \cap (\Gamma \setminus e)) = V(\Gamma) \cap \bar{r}$$

such that $e \subseteq \mathrm{ins}(C)$, that is, $r \subseteq \mathrm{ins}(C)$. Thus ins is even.

For the converse, suppose that $\mathscr{T}$ is not a tangle. By [4, Theorem (2.7)] there is an edge $e \in E(\Gamma)$ with $< \theta$ ends such that $(\Gamma \setminus e, K_e) \in \mathscr{T}$, with notation as before. Let r be the region of Γ' with $e \subseteq r$. For any circuit C of Γ' of length $< 2\theta$, $(\Gamma \cap \Delta, \Gamma \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$, where $\Delta = \mathrm{ins}(C)$, and so $(\Gamma \setminus e) \cup (\Gamma \cap \Delta) \neq \Gamma$, by the second tangle axiom; that is, $e \notin \Delta$, or equivalently, $r \not\subseteq \mathrm{ins}(C)$. Hence ins is not even. □

Another consequence of (6.1) is that, given a drawing in a connected surface, there is a 1-1 correspondence between its respectful tangles and the respectful tangles in its dual. We see this as follows. Let Γ' be a 2-cell drawing in a connected surface. One might ask, when is Γ' a radial drawing of some 2-cell drawing Γ with $E(\Gamma) \neq \emptyset$? First, Γ' must be bipartite, so let us assume that (A, B) is a bipartition of Γ'. Then it is easy to see that Γ' is a radial drawing of a 2-cell drawing Γ with $V(\Gamma) = A$ and $E(\Gamma) \neq \emptyset$ if and only if every region r of Γ' has perimeter of length 4. But if so, then there is another 2-cell drawing Γ^*, say, with $V(\Gamma^*) = B$, such that Γ' is a radial drawing of Γ^*. We call Γ^* the *dual* of Γ (it is unique in the usual sense of surface duals). Let $\theta \geq 1$ be an integer. Because there is a 1-1 correspondence between respectful tangles in Γ of order θ and even slopes in Γ' of order θ, because of (6.1) and (6.5), and also between respectful tangles in Γ^* of order θ and even slopes in Γ' of order θ, for the same reason, we deduce that there is a 1-1 correspondence between the respectful tangles of order θ in Γ and in Γ^*. In summary:

[F3] **Statement mistake.** As printed, the "if" part of (6.5) fails for order 2. Let Γ be the drawing in the sphere with one vertex v and one loop e. Its radial drawing Γ' joins v to the vertex of Γ' in each of the two discs bounded by e. So Γ' has no circuit, and its only region has perimeter of length 4. Hence for $\theta = 2$ there is no circuit or region of length $< 2\theta$, and the slope of order 2 is even. But the induced pretangle of order 2 in Γ contains $(\Gamma \setminus e, K_e)$, whose first part is the vertex v alone, and this violates the third axiom of a tangle. With the added hypothesis, (6.5) holds.

(6.6) *Let Γ, Γ^* be dual 2-cell drawings in a connected surface, with a common radial drawing Γ', and let $\theta \geq 1$ be an integer, such that if $\theta = 2$ then neither Γ nor Γ^* has a loop. There is a bijection $\mathscr{T} \leftrightarrow \mathscr{T}^*$ between respectful tangles $\mathscr{T}$ of order θ in Γ, and respectful tangles $\mathscr{T}^*$ of order θ in Γ^*, such that for each such $\mathscr{T}$, the slopes in Γ' derived* [p. 93] *from $\mathscr{T}$ and from $\mathscr{T}^*$ are equal. In particular, Γ has a tangle of order θ if and only if Γ^* does.*[F4]

We can also formulate (6.6) in terms of "dual" pairs of paintings, in the natural way. Incidentally, (6.6) contrasts with a result of [2], that dual graphs even in the sphere need not have the same tree-width. (A better example than that given in [2] is the cube and the octahedron; although they are dual in the sphere, one has tree-width 3 and the other 4.)

7. Some Topological Lemmas

Our object in the remainder of the paper is to study "planar neighbourhoods" of points, given a slope in a drawing on a surface. First, we shall need some lemmas about simply-connected sets.

(7.1) *If M is a triangulable connected 2-manifold, then any null-homotopic O-arc in M bounds a disc in M.*

This is [1, Theorem (1.7)].

(7.2) *If Σ is a connected surface, $Z \subseteq \Sigma$, and $F \subseteq Z$ is an O-arc null-homotopic in Z, then F bounds a disc $\Delta \subseteq Z$.*

Proof. Suppose that F bounds no such disc. From each of the (at most two) discs in Σ bounded by F, select a point of $\Sigma - Z$; and let X be the set of selected points. Then $|X| \leq 2$ and so $\Sigma - X$ is a triangulable connected 2-manifold, and F bounds no disc in $\Sigma - X$. By (7.1), F is not null-homotopic in $\Sigma - X$, and hence not null-homotopic in Z, since $Z \subseteq \Sigma - X$, a contradiction. □

A subset $Z \subseteq \Sigma$ is *nice* if there is a drawing Γ in Σ such that Z is the union of certain vertices, edges, and regions of Γ.

(7.3) *If $Z \subseteq \Sigma$ is nice and $\phi\colon S^1 \to Z$ is continuous and not null-homotopic in Z, then there is an O-arc $F \subseteq |\phi|$ which is not null-homotopic in Z.*

[F4] **Statement mistake.** The argument before (6.6) uses (6.5) in both Γ and Γ^*, so it needs the hypothesis of (6.5) in both. In the example of the note to (6.5), Γ^* has two vertices joined by one edge, and the pretangle of order 2 derived from the even slope is a tangle in Γ^* but not in Γ, so the bijection fails. With the added hypothesis, (6.6) holds.

Proof. For $Z = \Sigma$ this is [3, Theorem (11.2)]. The proof given there easily extends to nice sets Z; indeed, all that is needed is an open cover Δ_i $(i \in I)$ of Σ such that for $i, i' \in I$, every continuous $\psi\colon S^1 \to \Sigma$ with $|\psi| \subseteq (\Delta_i \cup \Delta_{i'}) \cap Z$ is null-homotopic in Z, and this clearly exists for nice sets Z. □

Incidentally, we are not sure if (7.3) holds without the "nice" requirement. From (7.2) and (7.3) we deduce

[p. 94] (7.4) *If Σ is connected and $X \subseteq \Sigma$ is nice, then X is simply-connected if and only if X is arc-wise connected and every O-arc $F \subseteq X$ bounds a disc in X.*

(7.5) *If Σ is connected, $X \subseteq \Sigma$ is closed and nice and $X \neq \Sigma$, and $Y \subseteq \Sigma$ is finite, and $X \cap Y = \emptyset$, and every O-arc in X is null-homotopic in $\Sigma - Y$, then there is a closed disc $\Delta \subseteq \Sigma - Y$ with $X \subseteq \Delta$.*

Proof. Choose disjoint closed discs D_y $(y \in Y)$ such that $X \cap D_y = \emptyset$ and $y \in D_y - bd(D_y)$ $(y \in Y)$. Let

$$\Sigma' = \Sigma - \bigcup_{y \in Y} (D_y - bd(D_y)).$$

Then Σ' is a compact 2-manifold with boundary. Every O-arc in X is null-homotopic in Σ', and so by [3, Theorem (11.10)] applied to Σ', there is a closed disc $\Delta \subseteq \Sigma'$ with $X \subseteq \Delta$. The result follows. □

(7.6) *If Σ is connected, and $X \subseteq \Sigma$ is closed, nice, and simply-connected, then $\Sigma - X$ is arc-wise connected.*

Proof. Let $a, b \in \Sigma - X$ be distinct. From (7.5) (with $Y = \{a, b\}$) there is a closed disc $\Delta \subseteq \Sigma - \{a, b\}$ with $X \subseteq \Delta$. But $\Sigma - \Delta$ is arc-wise connected and $X \cap (\Sigma - \Delta) = \emptyset$, and so a, b belong to the same arc-wise connected component of $\Sigma - X$. The result follows. □

(7.7) *If Σ is connected, $X, Y \subseteq \Sigma$ and $bd(X)$, $bd(Y)$ are both arc-wise connected, and $X \cap Y \neq \emptyset$, $X \cup Y \neq \Sigma$, and $bd(X) \cap bd(Y) = \emptyset$, then either $X \subseteq Y$ or $Y \subseteq X$.*

Proof. Since $\emptyset \neq X \cup Y \neq \Sigma$, there exists $z \in bd(X \cup Y)$. Since $bd(X \cup Y) \subseteq bd(X) \cup bd(Y)$ we may assume that $z \in bd(X)$. Since $bd(X) \cap bd(Y) = \emptyset$ it follows that $z \notin bd(Y)$. Since z belongs to the closure of $\Sigma - (X \cup Y)$ and hence to the closure of $\Sigma - Y$, it follows that $z \notin \bar{Y}$. Thus $bd(X) \not\subseteq Y$. Since $bd(X)$ is arc-wise connected and $bd(X) \cap bd(Y) = \emptyset$, it follows that $bd(X) \cap Y = \emptyset$. Since $X \cap Y \neq \emptyset$ we deduce that $Y \subseteq X$, as required. □

(7.8) *If $X, Y \subseteq \Sigma$ are closed and simply-connected and $X \cap Y \neq \emptyset$ is arc-wise connected, then $X \cup Y$ is simply-connected.*

Proof. Certainly $X \cup Y$ is arc-wise connected. By (3.1), any continuous $\phi\colon S^1 \to X \cup Y$ is homotopic in $X \cup Y$ to some continuous $\phi'\colon S^1 \to Y$; but ϕ' is null-homotopic in Y, and hence in $X \cup Y$, because Y is simply-connected. The result follows. □

[p. 95] (7.9) *If Σ is connected, $X, Y \subseteq \Sigma$ are simply-connected, and $X \cap Y \neq \emptyset$ is arc-wise connected and nice, and $X \cup Y \neq \Sigma$, then $X \cap Y$ is simply-connected.*

Proof. Let $z \in \Sigma - (X \cup Y)$. Let $F \subseteq X \cap Y$ be an O-arc. Then $F \subseteq X$ and X is simply-connected, and so by (7.2) F bounds a disc $\Delta_1 \subseteq X$. Similarly F bounds a disc $\Delta_2 \subseteq Y$. Now $z \notin \Delta_1, \Delta_2$ since $z \notin X, Y$, and yet there is at most one disc in Σ bounded by F not containing z, since Σ is connected. Thus $\Delta_1 = \Delta_2$, and $\Delta_1 \subseteq X \cap Y$. We deduce from (7.3) that $X \cap Y$ is simply-connected. □

(7.10) *If $X \subseteq \Sigma$ is arc-wise connected, $Y \subseteq \Sigma$ is closed, $bd(Y)$ is arc-wise connected, and $\emptyset \neq bd(Y) \subseteq X$, then $X \cap Y$ is arc-wise connected.*

Proof. Since $bd(Y)$ is arc-wise connected and non-empty, and $bd(Y) \subseteq X \cap Y$ since Y is closed, there is an arc-wise connected component C of $X \cap Y$ with $bd(Y) \subseteq C$. We claim that $C = X \cap Y$. For let $x \in X \cap Y$, let $y \in bd(Y)$, and let $\phi\colon [0,1] \to X$ be continuous with $\phi(0) = x$ and $\phi(1) = y$. Choose α with $0 \leq \alpha \leq 1$, maximum so that $\phi(\beta) \in Y$ for all β with $0 \leq \beta \leq \alpha$ (this is possible since Y is closed). Then $\phi(\alpha) \in bd(Y)$; for if $\alpha < 1$ this follows from the maximality of α, and if $\alpha = 1$ it is true since $y \in bd(Y)$. Moreover $\phi(\beta) \in X \cap Y$ for $0 \leq \beta \leq \alpha$, and so x and $\phi(\alpha)$ belong to the same arc-wise connected component of $X \cap Y$. Since $\phi(\alpha) \in bd(Y) \subseteq C$ it follows that $x \in C$. This proves our claim that $C = X \cap Y$, and so $X \cap Y$ is arc-wise connected. □

(7.11) *If $X \subseteq \Sigma$ is closed and arc-wise connected and $Y_1, \ldots, Y_r \subseteq \Sigma$ are closed discs with boundary in X, and $Z = X \cup Y_1 \cup \cdots \cup Y_r$, and every continuous $\phi\colon S^1 \to X$ is null-homotopic in Z, then every continuous $\phi\colon S^1 \to Z$ is null-homotopic in Z.*

Proof. For $0 \leq i \leq r$ let $Z_i = X \cup Y_1 \cup \cdots \cup Y_i$. We prove by induction on i that Z_i is arc-wise connected and every continuous $\phi\colon S^1 \to Z_i$ is null-homotopic in Z. Certainly this is true if $i = 0$. Suppose $1 \leq i \leq r$, and it holds for $i - 1$. By (7.10), since Z_{i-1} is arc-wise connected, Y_i is closed, $bd(Y_i)$ is arc-wise connected and $\emptyset \neq bd(Y_i) \subseteq Z_{i-1}$, it follows that $Z_{i-1} \cap Y_i$ is arc-wise connected. By (3.1), since Y_i is simply-connected and Z_{i-1}, Y_i are closed and $Z_{i-1} \cap Y_i \neq \emptyset$ is arc-wise connected, it follows that every continuous $\phi\colon S^1 \to Y_i \cup Z_{i-1}$ is homotopic in $Y_i \cup Z_{i-1}$ and hence in Z to

some $\phi\colon S^1 \to Z_{i-1}$, and hence is null-homotopic in Z. This completes the inductive argument, and the result follows. □

[p. 96] 8. Restraints

Throughout the remainder of the paper, Γ is a drawing in a connected surface Σ, with a slope ins of order $\theta \geq 1$, where 2θ is an integer. We say that $X \subseteq \Sigma$ is Γ-nice if X is a union of vertices, edges and regions of Γ.

Let $X \subseteq \Sigma$ with $X \neq \emptyset, \Sigma$, where X is closed, simply-connected and Γ-nice. By (7.5) there is a closed disc $\Delta \subseteq \Sigma$ with $X \subseteq \Delta$; and hence there is an open disc $\Delta' \subseteq \Sigma$ with $\Delta \subseteq \Delta'$. By (7.6), $\Delta' - X$ has a unique arc-wise connected component; and hence there is a closed walk of Γ tracing out $bd(X)$ in the natural sense. (Indeed, there are several such walks, obtained from one another by reinitializing and reversal.) Such a walk we call a perimeter walk of X. We denote the length of this walk by $l(X)$. Every edge of $\Gamma \cap bd(X)$ occurs in this walk, but some may occur twice, and so $l(X)$ may be greater than $|E(\Gamma \cap bd(X))|$. Indeed, its length equals the length of the perimeter of $\Sigma - X$ (as a region of $\Gamma \cap bd(X)$) defined before (6.5).

(8.1) *Let $X \subseteq \Sigma$ be closed, simply-connected, and Γ-nice with $X \neq \emptyset, \Sigma$. Let $v_0, e_1, v_1, \ldots, e_t, v_t = v_0$ be a perimeter walk of X. For $0 \leq i_1 \leq i_2 \leq i_3 \leq i_4 \leq t$, if $A, B \subseteq X$ are arc-wise connected and $v_{i_1}, v_{i_3} \in A$ and $v_{i_2}, v_{i_4} \in B$, then $A \cap B \neq \emptyset$.*

Proof. We may assume that $A = \{v_{i_1}\}$ if $v_{i_1} = v_{i_3}$, and A is a line with ends v_{i_1}, v_{i_3} otherwise; and similarly for B. Let $\Delta \subseteq \Sigma$ be an open disc with $X \subseteq \Delta$. Choose $u \in \Delta - X$, and for $1 \leq j \leq 4$ let F_j be a line in Δ such that

(i) F_j has ends u, v_{i_j} and $F_j \cap X = \{v_{i_j}\}$ $(1 \leq j \leq 4)$

(ii) for $1 \leq j < j' \leq 4$, $F_j \cap F_{j'} = \{u\} \cup (\{v_{i_j}\} \cap \{v_{i_{j'}}\})$, and so $F_1 \cup F_3 \cup A$ and $F_2 \cup F_4 \cup B$ are O-arcs

(iii) $F_1 \cup F_3 \cup A$ and $F_2 \cup F_4 \cup B$ cross at u.

Since O-arcs in a disc cannot cross only once, it follows that $A \cap B \neq \emptyset$, as required. □

If W is a closed walk and $\Gamma|W$ is confined, we define $\mathrm{ins}(W) = \mathrm{ins}(\Gamma|W)$. For convenient reference let us state:

(8.2) *Let W_1, W_2, W_3 be closed walks of Γ, each of length $< 2\theta$. Then there is a region r of Γ with $r \notin \mathrm{ins}(W_1) \cup \mathrm{ins}(W_2) \cup \mathrm{ins}(W_3)$.*

Proof. For $i = 1, 2, 3$, let Z_i be the set of regions of Γ included in $\mathrm{ins}(W_i)$. Then $\Gamma|W_i$ is confined and $N(Z_i) \subseteq \Gamma|W_i$, and so Z_i is small (with respect to the weighting $(V(\Gamma), 2\theta)$). The claim follows from (5.5). □

We say that $X \subseteq \Sigma$ is a *restraint* (of Γ, with respect to ins) if $X \neq \emptyset$ is closed, simply-connected and Γ-nice, $l(X) < 2\theta$, and $\Sigma - X \notin \mathrm{ins}(\Gamma \cap bd(X))$.

[p. 97] (8.3) *If $X \subseteq \Sigma$ is a restraint, then* $\mathrm{ins}(\Gamma \cap bd(X)) = X$.

Proof. By (7.6), $\Sigma - X$ is a region r_1 of $\Gamma \cap bd(X)$. But $\Gamma \cap bd(X)$ is confined, and so from (4.4), $\Sigma - \mathrm{ins}(\Gamma \cap bd(X))$ is a region r_2 of $\Gamma \cap bd(X)$. Since $\Sigma - X \notin \mathrm{ins}(\Gamma \cap bd(X))$ and hence $r_1 \cap r_2 \neq \emptyset$, it follows that $r_1 = r_2$, that is, $\mathrm{ins}(\Gamma \cap bd(X)) = X$, as required. □

(8.4) *If X_1, X_2, X_3 are restraints then $X_1 \cup X_2 \cup X_3 \neq \Sigma$.*

Proof. Let X_i have perimeter walk W_i ($i = 1, 2, 3$). By (8.3), $X_i = \mathrm{ins}(W_i)$ for $i = 1, 2, 3$, and the result follows from (8.2). □

(8.5) *Let W be a closed walk in Γ of length $< 2\theta$. Then* $\mathrm{ins}(W)$ *is a restraint and* $l(\mathrm{ins}(W)) \leq l(W)$.

Proof. Let $Z = \mathrm{ins}(W)$. Since $\Gamma|W$ is confined, it follows that any O-arc in $U(W)$ bounds a disc in Z, and hence, by (7.3), any continuous $\phi\colon S^1 \to U(W)$ is null-homotopic in Z. By repeated application of (3.1) it follows that Z is simply-connected. By (8.2), $Z \neq \Sigma$. Let us show that $l(Z) \leq l(W) < 2\theta$. Let W' be a perimeter walk of Z. Certainly every edge of W' is an edge of W. Moreover, if an edge e occurs twice in W' then $(\Gamma|W)\backslash e$ is disconnected (for otherwise e is in a circuit C of $\Gamma|W$, and $\mathrm{ins}(C)$ includes one of the two regions of Γ incident with e, which contradicts that e occurs twice in W') and so e occurs at least twice in W, because W is closed. Hence the length of W' is at most the length of W, and thus $l(Z) \leq l(W) < 2\theta$. Finally, we must show that $\Sigma - Z \notin \mathrm{ins}(\Gamma \cap bd(Z))$. But $\Gamma \cap bd(Z) \subseteq \Gamma|W$, and so

$$\mathrm{ins}(\Gamma \cap bd(Z)) \subseteq \mathrm{ins}(\Gamma|W) = \mathrm{ins}(W) = Z.$$

Since $Z \neq \Sigma$, it follows that $\Sigma - Z \notin \mathrm{ins}(\Gamma \cap bd(Z))$. Consequently Z is a restraint. □

(8.6) *If X, Y are restraints, and $X \cup Y$ is simply-connected and $l(X \cup Y) < 2\theta$, then $X \cup Y$ is a restraint.*

Proof. By (8.4), $X \cup Y \neq \Sigma$, and so $l(X \cup Y)$ is defined. Let $Z = \mathrm{ins}(\Gamma \cap bd(X \cup Y))$. By (8.5), Z is a restraint. By (8.4), $X \cup Y \cup Z \neq \Sigma$, and so

$$\Sigma - (X \cup Y) \notin \mathrm{ins}(\Gamma \cap bd(X \cup Y)).$$

But $X \cup Y$ is non-empty, closed, simply-connected and Γ-nice, and so $X \cup Y$ is a restraint. □

A restraint X is *extreme* if there is no restraint $Y \neq X$ with $X \subseteq Y$ and $l(Y) \leq l(X)$.

[p. 98] (8.7) *Let X be an extreme restraint, and let $a, b \in V(\Gamma \cap bd(X))$ be distinct. Let P be a walk in Γ from a to b, such that no edge or internal vertex of P is in X. Then there is a walk of $\Gamma \cap bd(X)$ from a to b which is shorter than P.*

Proof. We may assume that P is simple. Now by (7.6), $\Sigma - X$ is arc-wise connected. Since P is simple and has no internal vertex in X, it follows that $\Sigma - (X \cup U(P))$ has at most two arc-wise connected components, Δ_1 and Δ_2 say, where possibly $\Delta_1 = \Delta_2$. Since only two vertices of $\Gamma \cap bd(X)$ are incident with edges of P, and each of them is incident with only one edge of P, it follows that there are walks W_1, W_2 of $\Gamma \cap bd(X)$ from a to b such that $W_1 + W_2^{-1}$ is a perimeter walk of X and for $i = 1, 2$, $U(\Gamma|W_i) \subseteq \bar{\Delta}_i$.

We suppose for a contradiction that W_1, W_2 have length at least that of P. Thus $P + W_1^{-1}$ has length at most that of $W_2 + W_1^{-1}$, and hence at most $l(X)$. Let $X_1 = \mathrm{ins}(P + W_1^{-1})$; then X_1 is a restraint by (8.5). We claim that $\Delta_1 \subseteq X_1$. For there is a simple walk W from a to b with $\Gamma|W \subseteq \Gamma|W_1$, and hence $P + W^{-1}$ is a simple-closed walk of length $< 2\theta$. It follows that $\mathrm{ins}(P + W^{-1})$ includes one of Δ_1, Δ_2, and hence X_1 includes one of Δ_1, Δ_2. Suppose that $\Delta_2 \subseteq X_1$. Then $\bar{\Delta}_2 \subseteq X_1$, and therefore $U(\Gamma|W_2) \subseteq X_1$. Hence $bd(X) \subseteq X_1$. By (8.4), $X \cup X_1 \neq \Sigma$, and so $\Sigma - X_1 \not\subseteq X$. Now $\Sigma - X_1$ is arc-wise connected by (8.5) and (7.6), and is disjoint from $bd(X)$; and so $(\Sigma - X_1) \cap X = \emptyset$, that is, $X \subseteq X_1$. But $X \neq X_1$ since $U(P) \subseteq X_1$, and yet by (8.5) $l(X_1) \leq l(P + W_1^{-1}) \leq l(X)$, a contradiction since X is extreme. Thus $\Delta_2 \not\subseteq X_1$, and this proves our claim that $\Delta_1 \subseteq X_1$. Similarly there is a restraint X_2 with $\Delta_2 \subseteq X_2$; and so $X_1 \cup X_2 \cup X = \Sigma$, contrary to (8.4). Thus our supposition that W_1, W_2 both have length at least that of P was false, and the result follows. □

(8.8) *If X_1, X_2 are extreme restraints and $X_1 \cap X_2 \neq \emptyset$ then $X_1 \cap X_2$ and $X_1 \cup X_2$ are simply-connected.*

Proof. We shall first prove that $X_1 \cap X_2$ is arc-wise connected. Let its arc-wise connected components be $D_1, \ldots, D_k$. We may assume that

(1) $D_h \cap bd(X_1) \neq \emptyset$, for $1 \leq h \leq k$.

Subproof. We may assume that $D_h \neq X_2$, for otherwise $X_2 \subseteq X_1$, $k = 1$ and we are finished. Choose $a \in D_h$ and $b \in X_2 - D_h$, and let ϕ be a curve with $\phi(0) = a$, $\phi(1) = b$ and $|\phi| \subseteq X_2$. Choose α with $0 \leq \alpha \leq 1$ maximal so that $\phi(\beta) \in X_1$ for $0 \leq \beta \leq \alpha$ (this is possible since X_1 is closed). Then $\alpha < 1$, and so $\phi(\alpha) \in bd(X_1)$; and moreover, $\phi(\beta) \in X_1 \cap X_2$ for $0 \leq \beta \leq \alpha$, and so $\phi(\alpha) \in D_h$. This proves (1).

[p. 99] Let $v_0, e_1, v_1, e_2, \ldots, e_t, v_t = v_0$ be a perimeter walk of X_1. For

$1 \le h \le k$ let

$$i(h) = \min\{i : 0 \le i \le t, v_i \in D_h\}$$
$$j(h) = \max\{j : 0 \le j \le t, v_j \in D_h\}.$$

Both $i(h)$ and $j(h)$ exist by (1). Choose h with $1 \le h \le k$ such that $j(h) - i(h)$ is minimum.

(2) *If* $0 \le i, j \le t$ *and* $i(h) \le i \le j(h)$ *and either* $j < i(h)$ *or* $j > j(h)$, *then* $v_i \ne v_j$, *and none of* $D_1, \ldots, D_k$ *contains both* v_i *and* v_j.

Subproof. Since either $j < i(h)$ or $j > j(h)$ it follows that $v_j \notin D_h$. If $i(h) \le i \le j(h)$ and $v_i = v_j$, let $D = \{v_j\}$; and if $i(h) \le i \le j(h)$ and v_i, v_j both belong to some $D_{h'}$, let $D = D_{h'}$. In the latter case $h' \ne h$ since $v_j \notin D_h$, and so in either case $D \cap D_h = \emptyset$, contrary to (8.1). This proves (2).

(3) *For all* $h' \ne h$ *with* $1 \le h' \le k$, $v_i \notin D_{h'}$ *for* $i(h) \le i \le j(h)$.

Subproof. From the minimality of $j(h) - i(h)$, either $i(h') \le i(h)$ or $j(h') \ge j(h)$, and by the symmetry we may assume the latter without loss of generality. Since $h' \ne h$ and hence $D_h \cap D_{h'} = \emptyset$, it follows that $j(h') \ne j(h)$ and so $j(h') > j(h)$. The claim follows from (2) (setting $j = j(h')$).

Let $i = i(h)$ and $j = j(h)$. We may assume that

(4) $i \ge 1$ *and* $j \le t - 1$ *and* $e_i, e_{j+1} \subseteq \Sigma - X_2$.

Subproof. If $i = 0$ or $j = t$ then $v_0 = v_t \in D_h$ and $i(h) = 0$ and $j(h) = t$; and then by (1) and (3), $k = 1$ and we are finished. Thus we may assume that $i \ge 1$ and $j \le t - 1$. Since $v_{i-1} \notin D_h$ it follows that $e_i \not\subseteq D_h$ (because D_h is closed) and similarly $e_{j+1} \not\subseteq D_h$. Thus $e_i, e_{j+1} \subseteq \Sigma - X_2$. This proves (4).

Let $Y_1 = \{v_i, v_{i+1}, \ldots, v_j\}$, $Y_2 = \{v_0, v_1, \ldots, v_{i-1}, v_{j+1}, v_{j+2}, \ldots, v_t\}$. By (2),

(5) $Y_1 \cap Y_2 = \emptyset$, *and no component of* $X_1 \cap X_2$ *meets both* Y_1 *and* Y_2.

Let $Z_i = bd(X_2) \cap Y_i$ $(i = 1, 2)$. Then $Z_1 \cup Z_2 = bd(X_1) \cap bd(X_2) \cap V(\Gamma)$. Since $v_i, v_j \in D_h \subseteq X_2$ and $e_i, e_{j+1} \subseteq \Sigma - X_2$ by (4), it follows that $v_i, v_j \in Z_1$ and so $Z_1 \ne \emptyset$. Suppose that also $Z_2 \ne \emptyset$. For $i = 1, 2$ let Q_i be the shortest walk of $\Gamma \cap bd(X_i)$ from a vertex in Z_1 to a vertex in Z_2. These exist, for $Z_1, Z_2 \ne \emptyset$ and $bd(X_1), bd(X_2)$ are both arc-wise connected. Q_1 and Q_2 are both simple walks, and no internal vertex of either belongs to $Z_1 \cup Z_2$. Let Q_1, Q_2 have ends a_1, b_1 and a_2, b_2, respectively. From (5), $U(Q_1), U(Q_2) \not\subseteq X_1 \cap X_2$, and so $U(Q_1) \not\subseteq X_2$ and $U(Q_2) \not\subseteq X_1$. Since no internal vertex of Q_1 belongs to $bd(X_2)$ we deduce that $U(Q_1) \cap X_2 =$ [p. 100] $\{a_1, b_1\}$, and similarly $U(Q_2) \cap X_1 = \{a_2, b_2\}$. From (8.7) applied to X_1 it follows that some walk of $\Gamma \cap bd(X_1)$ between a_1 and b_1 is shorter than Q_2

and hence Q_1 is shorter than Q_2, because Q_1 is the shortest such walk. But similarly, by (8.7) applied to X_2, Q_2 is shorter than Q_1, a contradiction. Thus $Z_2 = \emptyset$, and hence $bd(X_2) \cap Y_2 = \emptyset$.

Since $e_1 \subseteq \Sigma - X_2$, it follows that $Y_2 \subseteq \Sigma - X_2$, and hence no component of $X_1 \cap X_2$ intersects Y_2. By (1) and (3), $k = 1$ and so $X_1 \cap X_2$ is arc-wise connected. From (7.8), $X_1 \cup X_2$ is simply-connected. Since $X_1 \cup X_2 \neq \Sigma$ by (8.4), it follows that $X_1 \cap X_2$ is simply-connected by (7.9). □

Let $a, b \in \Sigma$. A restraint X is *a, b-optimal* if $a, b \in X$, and there is no restraint Y with $a, b \in Y$ and $l(Y) < l(X)$. It is *a, b-extreme* if it is a, b-optimal and extreme. If b is an edge or region, we say that X is *a, b-optimal* if it is a, b'-optimal for some $b' \in b$. (We observe that if $b' \in b$, then $b' \in X$, where X is a restraint if and only if $b \subseteq X$.) "a, b-extreme" is generalized similarly.

(8.9) *Let $a, b \in \Sigma$, let X' be an a, b-optimal restraint, and let X be an extreme restraint with $a, b \in X$. Then $X' \subseteq X$.*

Proof. We may assume that X' is a, b-extreme. By (8.8) $X \cap X'$ and $X \cup X'$ are simply-connected, and $X \cup X' \neq \Sigma$ by (8.4). Let W, W', W_1, W_2 be perimeter walks of X, X', $X \cap X'$, $X \cup X'$, respectively.

(1) $l(X \cap X') \geq l(X')$.

Subproof. We may assume that $l(X \cap X') < 2\theta$, and so $\Sigma - (X \cap X') \not\subseteq \text{ins}(W_1)$ by (8.2), since $X \cap X' \subseteq X = \text{ins}(W)$ by (8.3). It follows that $X \cap X'$ is a restraint. But $a, b \in X \cap X'$, and the claim follows since X' is a, b-optimal.

(2) $l(X \cup X') + l(X \cap X') \leq l(X) + l(X')$.

Subproof. Any edge of Γ contributes at least as much to the right side of this inequality as it does to the left.

We deduce from (1) and (2) that $l(X \cup X') \leq l(X)$. By (8.6), $X \cup X'$ is a restraint. Since $l(X \cup X') \leq l(X)$ and X is extreme, we deduce that $X \cup X' = X$, that is, $X' \subseteq X$, as required. □

(8.10) *Let $c \in \Sigma$, let $0 \leq k < 2\theta$, and let Z be the union of all restraints X with $l(X) \leq k$ and $c \in X$. Then Z is simply-connected.*

[p. 101] *Proof.* First, we assume that Γ is connected. Choose $u \in V(\Gamma)$ such that

$u = c$	if $c \in V(\Gamma)$
$u \in \bar{e}$	if $c \in e$ for some edge e of Γ
$u \in \bar{r}$	if $c \in r$ for some region r of Γ.

For each $v \in V(\Gamma) \cap Z$, let X_v be a c, v-optimal restraint. Since $v \in Z$ it follows that $l(X_v) \leq k$, and hence $X_v \subseteq Z$. Since $\Gamma \cap bd(X_v)$ is connected

and so is Γ, and $u, v \in X_v$, there is a simple walk P_v of Γ from v to u with $U(P_v) \subseteq X_v$.

(1) *For any edge $e \in E(\Gamma)$ with $e \subseteq Z$ and with ends v_1, v_2, let P_e be the walk v_1, e, v_2; then P_e is homotopic in Z to $P_{v_1} + P_{v_2}^{-1}$.*

Subproof. Let X be a c, e-extreme restraint. Since $e \subseteq Z$ it follows that $l(X) \leq k$, and hence $X \subseteq Z$. Now $v_1 \in X$ and so $X_{v_1} \subseteq X$ by (8.9). Thus P_{v_1} is a walk of $\Gamma \cap X$, and similarly so is P_{v_2}. Since X is simply-connected and $X \subseteq Z$, and P_e, $P_{v_1} + P_{v_2}^{-1}$ are both walks of $\Gamma \cap X$ from v_1 to v_2, the claim follows.

(2) *Any closed walk of $\Gamma \cap Z$ is null-homotopic in Z.*

Subproof. Let $v_0, e_1, v_1, \ldots, e_t, v_t = v_0$ be a closed walk of $\Gamma \cap Z$. By (1) it is homotopic in Z to

$$P_{v_0} + P_{v_1}^{-1} + P_{v_1} + P_{v_2}^{-1} + \cdots + P_{v_{t-1}} + P_{v_0}^{-1};$$

and since each $P_{v_i}^{-1} + P_{v_i}$ is null-homotopic in Z, the claim follows.

If X is a restraint with $l(X) \leq k$ and $c \in X$, then by (8.3), X is the union of $bd(X)$ and finitely many discs with boundary in $bd(X)$ (namely, all discs of the form $\mathrm{ins}(C)$, where $C \subseteq \Gamma \cap bd(X)$ is a circuit of length $< 2\theta$). Since $bd(X) \subseteq U(\Gamma \cap Z)$, it follows that Z is the union of $U(\Gamma \cap Z)$ with some closed discs with boundary in $U(\Gamma \cap Z)$; but $U(\Gamma \cap Z)$ is arc-wise connected, and so Z is simply-connected, from (2), (7.3), and (7.11). This completes the proof when Γ is connected.

For the general case, let Γ have components $\Gamma_1, \ldots, \Gamma_t$, say. For $1 \leq i \leq t$ let Z_i be the union of all restraints X with $l(X) \leq k$ and $c \in X$ such that $bd(X) \subseteq U(\Gamma_i)$. Then $Z = Z_1 \cup \cdots \cup Z_t$, since every restraint has connected boundary. Let $1 \leq i < j \leq t$. We claim that either $Z_i \subseteq Z_j$ or $Z_j \subseteq Z_i$. For suppose not, and let $a \in Z_i - Z_j$, $b \in Z_j - Z_i$. Let X be a restraint with $c, a \in X$, $l(X) \leq k$ and $bd(X) \subseteq U(\Gamma_i)$, and define Y similarly with $c, b \in Y$ and $bd(Y) \subseteq U(\Gamma_j)$. Then $bd(X) \cap bd(Y) = \emptyset$, and $X \cap Y \neq \emptyset$, and $X \cup Y \neq \Sigma$ by (8.4), and $X \not\subseteq Y$ (since $a \in X$, and $a \notin Z_j \supseteq Y$) and $Y \not\subseteq X$ similarly. But this contradicts (7.7). Thus either $Z_i \subseteq Z_j$ or $Z_j \subseteq Z_i$ for $1 \leq i < j \leq t$. Hence there exists i with $1 \leq i \leq t$ such that $Z_1 \cup \cdots \cup Z_t = Z_i$, and hence $Z = Z_i$.

[p. 102] Let ins' be the restriction of ins to the set of circuits of Γ_i of length $< 2\theta$. This is a slope of order θ in Γ_i, by (4.2). Let us say that X is an ins'-restraint if it is a restraint of Γ_i with respect to ins'. Then X is an ins'-restraint if and only if X is a restraint of Γ with respect to ins and $bd(X) \subseteq U(\Gamma_i)$. But, by the argument above when Γ is connected, the union of ins'-restraints X with $l(X) \leq k$ and $c \in X$ is simply-connected. This union equals Z_i, and $Z_i = Z$, and so Z is simply-connected. □

(8.11) *Let $c \in \Sigma$, let $0 \le k < 2\theta$, and let Z be the union of all restraints X with $l(X) \le k$ and $c \in X$. Then for any restraint Y with $c \in Y$, if $bd(Y) \subseteq Z$ then $Y \subseteq Z$.*

Proof. Suppose that Y is a restraint with $c \in Y$, $bd(Y) \subseteq Z$ and $Y \not\subseteq Z$. Since $\Sigma - Z$ is arc-wise connected, by (7.6) and (8.10), and $bd(Y) \cap (\Sigma - Z) = \emptyset$, it follows that $\Sigma - Z \subseteq Y$. Let X be a restraint with $Y \subseteq X$ and $l(X) \le l(Y)$, chosen with X maximal; then X is extreme, $c \in X$, and $\Sigma - Z \subseteq X$ and hence $bd(X) \subseteq Z$. Let Z' be the union of all restraints X' with $l(X') \le k$, $c \in X'$ and $bd(X') \subseteq X$.

Suppose that $Z' \not\subseteq X$. Then there is a restraint X' with $l(X') \le k$, $c \in X'$, $bd(X') \subseteq X$, and $X' \not\subseteq X$. It follows, since $\Sigma - X$ is arc-wise connected by (7.6), that $X \cup X' = \Sigma$, contrary to (8.4). Hence $Z' \subseteq X$, and we claim that $Z' = Z \cap X$. Since certainly $Z' \subseteq Z$, it follows that $Z' \subseteq Z \cap X$. On the other hand, let $z \in Z \cap X$, and let X' be a c, z-optimal restraint. Then $l(X') \le k$. Since $c, z \in X$ it follows from (8.9) that $X' \subseteq X$, and so $bd(X') \subseteq X$. Thus $X' \subseteq Z'$, and so $z \in Z'$. We deduce that $Z \cap X \subseteq Z'$, and so $Z' = Z \cap X$.

Now $\Sigma - X$, $\Sigma - Z$ are non-empty open sets, and they are disjoint, since $\Sigma - Z \subseteq X$. Hence $(\Sigma - X) \cup (\Sigma - Z)$ is not arc-wise connected. By (7.6), $X \cap Z = Z'$ is not simply-connected. But this contradicts (8.10) applied to $\Gamma \cap X$. We deduce that there is no such Y, as required. □

(8.12) *If $c \in \Sigma$, there is a region r of Γ such that there is no restraint X with $c \in X$ and $r \subseteq X$.*

Proof. The union of all restraints X with $l(X) = 0$ and $c \in X$ is $\{c\}$ if $c \in V(\Gamma)$ and $\emptyset$ otherwise. Thus we may choose $k \ge 0$ maximum such that $k < 2\theta$ and $Z_k \ne \Sigma$, where Z_k is defined to be the union of all restraints X with $l(X) \le k$ and $c \in X$. Since Z_k is closed, there is a region $r \subseteq \Sigma - Z_k$, and we claim that r satisfies the theorem. For suppose not. Then $k \le 2\theta - 2$ and $Z_{k+1} = \Sigma$, and in particular there is a restraint X with $l(X) \le k + 1$ and $c \in X$ and $r \subseteq X$. Choose such a restraint, extreme. Since $X \not\subseteq Z_k$ it follows from (8.11) that $bd(X) \not\subseteq Z_k$. Let $b \in bd(X) - Z_k$. Since $b \in bd(X)$ and X is closed, there is a region r' of Γ with $r' \subseteq \Sigma - X$ and $b \in \bar{r}'$. Since $b \notin Z_k$ it follows that $r' \cap Z_k = \emptyset$. Since $Z_{k+1} = \Sigma$, there is a c, r'-optimal restraint X', and $l(X') = k + 1$. Since $b \notin Z_k$ and $b \in X'$ it follows that X' is [p. 103] c, b-optimal. Since $c, b \in X$ and $X' \not\subseteq X$ (because $r' \subseteq X'$ and $r' \not\subseteq X$) this contradicts (8.9). Thus r satisfies the theorem, as required. □

Statement (8.12) turns out to be quite significant. For instance, the result of [5], that for planar graphs branch-width is computable in polynomial time, relies heavily on (8.12), and it will also have several applications in

future papers of this series.

Another consequence of (8.9) is the following.

(8.13) *Let $a, b \in \Sigma$ with $\{a, b\} \cap U(\Gamma) \subseteq V(\Gamma)$; and let X be a restraint with $a, b \in X$. Suppose that there is no restraint Y with $X \subseteq Y$ and $l(Y) < l(X)$. Then X includes an a, b-optimal restraint.*

Proof. We have a slope ins in Γ of order θ. Let Γ' be obtained from Γ by subdividing each edge $e \in E(\Gamma)$ $2\theta - 1$ times if $e \subseteq X$, and 2θ times otherwise. (Thus, e corresponds to a path of Γ' with 2θ or $2\theta + 1$ edges.) Let C be a circuit of Γ, and C' the circuit of Γ' with $U(C) = U(C')$. Then

$$2\theta|E(C)| \leq |E(C')| \leq (2\theta + 1)|E(C)|$$

and so C has length $< 2\theta$ if and only if C' has length $< 4\theta^2$. If C has length $< 2\theta$, we define $\mathrm{ins}'(C') = \mathrm{ins}(C)$. Then ins' is a slope in Γ' of order $4\theta^2$.

Define $\mathscr{R}$ to be the set of restraints of Γ with respect to ins, and $\mathscr{R}'$ to be the set of restraints of Γ' with respect to ins' (replacing 2θ by $4\theta^2$ in the definition of "restraint"). Let us say that $X' \in \mathscr{R}'$ is *smooth* if every vertex with valency ≤ 1 of $\Gamma'|bd(X')$ belongs to $V(\Gamma)$. Then a computation similar to that above yields that $\mathscr{R} \subseteq \mathscr{R}'$, and $\mathscr{R}$ contains precisely the smooth members of $\mathscr{R}'$.

Now $X \in \mathscr{R}$ and $l'(X) = 2\theta l(X)$, where l' denotes the perimeter length of X in Γ'. For any $Y \in \mathscr{R}'$ with $X \subseteq Y \neq X$, we claim that $l'(Y) > l'(X)$. For if Y is smooth then $Y \in \mathscr{R}$; and since some edge of the perimeter of Y is not included in X, and since from the hypothesis $l(Y) \geq l(X)$, it follows that $l'(Y) > 2\theta l(Y) \geq 2\theta l(X) = l'(X)$ as claimed. On the other hand, if Y is not smooth then, since X is smooth, there exists a smooth $Y' \in \mathscr{R}'$ with $X \subseteq Y' \subseteq Y$ and with $l'(Y') < l'(Y)$. But then $l'(Y') \geq l'(X)$, since either $Y' = X$ or $l'(Y') > l'(X)$. Thus $l'(Y) > l'(Y') \geq l'(X)$. This proves our claim. In particular, X is extreme (in Γ' with respect to ins'). Let $X' \in \mathscr{R}'$ be an a, b-optimal restraint (in Γ', with respect to ins'). By (8.9), $X' \subseteq X$. Moreover, X' is smooth, since $\{a, b\} \cap U(\Gamma) \subseteq V(\Gamma)$ and X' is a, b-optimal; and hence $X' \in \mathscr{R}$. We claim that X' is a, b-optimal in Γ with respect to ins. For suppose not, and let $Y \in \mathscr{R}$ with $a, b \in Y$ and $l(Y) < l(X')$. Then $Y \in \mathscr{R}'$, and

$$\begin{aligned} l'(Y) &\leq (2\theta + 1)l(Y) \leq (2\theta + 1)(l(X') - 1) \\ &= 2\theta l(X') + (l(X') - 2\theta - 1) < 2\theta l(X') \leq l'(X') \end{aligned}$$

[p. 104] since $l(X') \leq 2\theta$, contrary to the fact that X' is a, b-optimal in Γ' with respect to ins'. Hence X' is a, b-optimal in Γ with respect to ins, and the result follows. □

9. The Distance Function

The results of the last section suggest a natural distance function on Σ. Our final objective is to establish some basic properties of this distance; we shall pursue it further in another paper. The sets $\{v\}$ $(v \in V(\Gamma))$ and the edges and regions of Γ are called the *atoms* of Γ; the set of atoms is denoted by $A(\Gamma)$.

For $a, b \in A(\Gamma)$, let us define $d(a, b)$ as follows:

(i) if $a = b$ then $d(a, b) = 0$

(ii) if $a \neq b$ and $a, b \subseteq X$ for some restraint X, we define $d(a, b)$ to be $\min \frac{1}{2}l(X)$, taken over all such X

(iii) if (i) and (ii) do not apply then $d(a, b) = \theta$.

We observe that, if (ii) applies, then $d(a, b)$ is $\frac{1}{2}l(X)$, where X is an a, b-extreme restraint (by which we mean an a', b'-extreme restraint for some $a' \in a$ and $b' \in b$). In general $d(a, b)$ may be non-integral; but in our applications Γ will be a radial drawing of some painting and hence bipartite, and in that case $d(a, b)$ will be integral.

(9.1) *If* $a, b, c \in A(\Gamma)$ *then* $d(a, b) + d(b, c) \geq d(a, c)$.

Proof. We may assume that $d(a, b) < d(a, c)$, and so $b \neq c$, and similarly $a \neq b$. Since $d(a, c) \leq \theta$, we may also assume that $d(a, b) + d(b, c) < \theta$, and so $d(a, b) = \frac{1}{2}l(X)$, $d(b, c) = \frac{1}{2}l(Y)$, where X, Y are a, b-extreme and b, c-extreme restraints, respectively. Since $X \cap Y \neq \emptyset$, it follows from (8.8) that $X \cup Y$ is simply-connected. Moreover, $l(X \cup Y) \leq l(X) + l(Y) < 2\theta$, and so $X \cup Y$ is a restraint by (8.6). Hence

$$d(a, c) \leq \frac{1}{2}l(X \cup Y) \leq \frac{1}{2}l(X) + \frac{1}{2}l(Y) = d(a, b) + d(b, c)$$

as required. □

Let X be a restraint such that either

(i) $|X| = 1$, or

(ii) X is a line, or

(iii) X is a closed disc, or

(iv) $X = D_1 \cup D_2$, where D_1, D_2 are closed discs with $|D_1 \cap D_2| = 1$,
[p. 105] or

(v) $X = D \cup F$, where D is a closed disc and F is a line with one end in D and no other point in D, or

(vi) $X = D_1 \cup D_2 \cup F$, where D_1, D_2 are disjoint closed discs and F is a line with one end in D_1, the other in D_2, and no other point in $D_1 \cup D_2$.

Such a restraint X we call a *tie*. Our next result implies that in defining our distance function d we could confine ourselves to ties.

(9.2) *Let $a, b \in A(\Gamma)$ and let Y be a restraint with $a, b \subseteq Y$. Then there is a tie X with $a, b \subseteq X \subseteq Y$ and $l(X) \leq l(Y)$.*

Proof. If there is a closed disc $D \subseteq Y$ with $\mathrm{bd}(D) \subseteq \mathrm{bd}(Y)$ and $a \subseteq D$, let $A = D$. Otherwise, $a \subseteq \mathrm{bd}(Y)$, since $Y = \mathrm{ins}(\Gamma \setminus \mathrm{bd}(Y))$, by (8.3); in this case let $A = \bar{a}$. We observe that in this second case, if $a \in E(\Gamma)$ then no region incident with a is in Y (again, because $Y = \mathrm{ins}(\Gamma \setminus \mathrm{bd}(Y))$), and so a occurs twice in the perimeter walk of Y. Let $A^* = A - \mathrm{bd}(A)$. (Thus $A^* = \emptyset$ unless A is a disc.) Define B, B^* similarly.

Now $\mathrm{bd}(A) \cap B^* = \emptyset$ because $\mathrm{bd}(A) \subseteq \mathrm{bd}(Y)$ and $B^* \cap \mathrm{bd}(Y) = \emptyset$, and similarly $\mathrm{bd}(B) \cap A^* = \emptyset$. If $A^* \cap B^* \neq \emptyset$, then since A^* is arc-wise connected and $A^* \cap \mathrm{bd}(B^*) = \emptyset$ it follows that $A^* \subseteq B^*$, and similarly $B^* \subseteq A^*$. But then $A = B$ and A is a tie with $a, b \subseteq A$ and $l(A) \leq l(Y)$, as required. Thus we may assume that $A^* \cap B^* = \emptyset$, and so $A \cap B = \mathrm{bd}(A) \cap \mathrm{bd}(B)$. Suppose that $|\mathrm{bd}(A) \cap \mathrm{bd}(B)| \geq 2$. Then neither A nor B is a singleton, and

$$\Gamma' = (\Gamma \cap \mathrm{bd}(A)) \cup (\Gamma \cap \mathrm{bd}(B))$$

is non-separable and confined. If Γ' has a circuit then by (4.3)(iii) there is a circuit C of Γ' with $\mathrm{ins}(\Gamma') = \mathrm{ins}(C)$ and hence with

$$a, b \subseteq A \cup B \subseteq \mathrm{ins}(\Gamma') = \mathrm{ins}(C)$$

and

$$\mathrm{bd}(\mathrm{ins}(C)) = U(C) \subseteq U(\Gamma') \subseteq \mathrm{bd}(A) \cup \mathrm{bd}(B) \subseteq \mathrm{bd}(Y);$$

and so $\mathrm{ins}(C)$ is the required tie, for $|E(C)| \leq |E(\Gamma')| \leq l(Y)$. If Γ' has no circuit, then $A = B = \bar{e}$ for some edge e occuring twice in the perimeter of Y, and $X = \bar{e}$ is the desired tie. We may assume then that $|\mathrm{bd}(A) \cap \mathrm{bd}(B)| \leq 1$, and so $|A \cap B| \leq 1$. If $|A \cap B| = 1$ then $A \cup B$ is the required tie, and so we may assume that $A \cap B = \emptyset$.

Let $v_0, e_1, v_1, \ldots, e_t, v_t = v_0$ be a perimeter walk of Y, where $v_0 \in A$. There exists i such that $v_i \in B$. Thus we may choose $0 \leq i_1 \leq i_2 \leq i_3 \leq i_4 \leq t$ such that $v_{i_1} \in A$, $v_{i_2} \in B$, $v_{i_3} \in B$, $v_{i_4} \in A$, and $i_2 - i_1 + i_4 - i_3$ is minimum. Let W_1 [p. 106] be the walk $v_{i_1}, e_{i_1+1}, \ldots, v_{i_2}$ and let W_2 be the walk $v_{i_3}, e_{i_3+1}, \ldots, v_{i_4}$. Then no internal vertex of W_1 or W_2 lies in $A \cup B$, and no edge of W_1 or W_2 lies in A or in B. Hence the sum of the lengths of W_1 and W_2 is at most $l(Y) - \alpha - \beta$, where α is the number of values of $i (1 \leq i \leq t)$ with $e_i \subseteq \mathrm{bd}(A)$, and β is defined similarly. We may assume then that W_1 has length $\leq \frac{1}{2}(l(Y) - \alpha - \beta)$. Now $v_{i_1} \neq v_{i_2}$ since $A \cap B = \emptyset$; choose a line F with ends v_{i_1}, v_{i_2} and with $F \subseteq U(W_1)$. Then $X = A \cup B \cup F$ is the required tie; for if γ is the number of edges of Γ included in F, then $\gamma \leq \frac{1}{2}(l(Y) - \alpha - \beta)$ and $l(X) = \alpha + \beta + 2\gamma$, and so $l(X) \leq l(Y)$. □

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